Does Per-Frequency-Band Graph Learning Help Copper Price Forecasting? A Rigorous Evaluation and a Diagnosis of Representation Collapse

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***Abstract*—We build a leakage-safe, capacity-matched protocol testing whether VMD-MFGNN, a per-frequency-band graph neural network, improves copper futures forecasting. Across seven models and four horizons on one 2010–2025 chronological split, none beats a zero-forecast baseline on mean absolute error, and it also wins RMSE at two horizons. We then diagnose why VMD-MFGNN's per-band graphs show no advantage over a pooled alternative. Trained checkpoints reveal the graph embeddings collapsed isotropically, RMS magnitude falling 32-fold, while two independently trained models share embedding directions at cosine similarity above 0.99. Training rescaled the embeddings without ever rotating them toward task-relevant structure. We attribute this to weight decay overwhelming a weak task gradient, since a near-uniform adjacency already suffices. A targeted fix, normalizing embeddings and excluding them from weight decay, mechanically resolves the collapse. The deeper problem persists: corrected embeddings move only 0.6–2.0% from random initialization, so the adjacency stays statistically indistinguishable from an untrained graph. Separately, several models, including a tuned re-run that first appeared to reverse our negative finding, emitted near-constant forecasts whose directional accuracy matched the test period's majority-class frequency. We retract that apparent reversal once we identified the artifact. We report this as a general finding: directional accuracy is uninterpretable without checking prediction sign concentration and a majority-class null alongside it. Our contribution is therefore an evaluation protocol, not a winning architecture. It includes a disclosed naive-baseline comparison this literature usually omits, a quantified graph-collapse failure mode relevant to low-node-count financial GNNs, and a warning about what directional accuracy alone can conceal.**

***Index Terms*—copper price forecasting, graph neural networks, variational mode decomposition, multi-frequency analysis, commodity markets.**

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I. Introduction

Copper is among the most economically significant industrial metals, colloquially termed "Dr. Copper" in financial commentary for its perceived ability to signal macroeconomic health. Global copper output reached approximately 22.8 million metric tonnes in 2024, with China consuming over 50% of global demand (IEA, 2024). The ongoing energy transition—electric vehicles require 53–83 kg of copper per unit versus 15–23 kg for internal combustion engines—is creating structural demand shifts that complicate traditional forecasting approaches (IEA, 2024). Forecasting errors translate directly into hedging and inventory decisions across the copper supply chain: mine operators and smelters price forward sales off short-horizon forecasts, while EV and grid-infrastructure manufacturers plan multi-year procurement off longer-horizon ones, so both the accuracy and the honesty of a forecasting method's reported accuracy matter in a way that is specific to this market, not generic to commodities.

Recent advances in copper price forecasting have progressed along two largely overlapping tracks. The first track employs signal decomposition methods, particularly Variational Mode Decomposition (VMD), to separate copper price series into frequency components before forecasting each independently. VMD-LSTM (Liu et al., 2020) established the benchmark and is, as of this writing, the most-cited paper in this specific sub-area (290 citations per Google Scholar, accessed 2026-08), and secondary decomposition approaches (e.g., CEEMDAN+VMD) represent the current frontier (Zhang et al., 2025a; Zhang and Ke, 2025). Most of these methods treat copper as an isolated univariate series, discarding the inter-commodity dependencies that drive prices, although a small number of very recent papers now combine VMD with graph neural networks (Section II).

The second track applies Graph Neural Networks (GNN) to model cross-asset relationships. GNNs have shown strong results in stock prediction (Cheng et al., 2022; Qian et al., 2024) and have recently been applied to copper via a Multi-view Graph Transformer (Sun et al., 2025), which constructs a single graph over undecomposed prices. Separately, at least one recent paper, MBTI-Net (Li and Liu, 2026), combines VMD decomposition with a GNN for copper futures, but pools the decomposed modes into a single graph rather than constructing one graph per frequency band. In both cases, a single graph structure collapses whatever multi-scale dependency structure exists across frequency bands into one static representation.

The premise motivating a per-band graph is intuitive and, on its face, plausible: commodity relationships are plausibly multi-scale, so the copper-dollar correlation at low frequencies (reflecting long-term macroeconomic co-movement) may differ structurally from the copper-dollar relationship at high frequencies (reflecting daily trading noise), and a single graph, pooled or otherwise, cannot represent both simultaneously. **This paper does not assume that premise; it tests it.** We built VMD-MFGNN, a Multi-Frequency Graph Neural Network that constructs a *separate* learned inter-variable graph for each VMD frequency band, exactly the architectural distinction MBTI-Net-style pooling forgoes, and evaluate it against six baselines and a matched pooled-graph ablation under a leakage-safe, capacity-disclosed protocol. We report what we found even where it complicates the paper's own premise: none of the seven models tested, including VMD-MFGNN, beats a trivial zero-forecast baseline on mean absolute error at any horizon, and direct inspection of the trained model's learned graphs shows the per-band mechanism did not learn differentiated relational structure at all—its embeddings collapsed toward a degenerate, near-uniform solution during training, a failure mode we diagnose quantitatively in Section V-E1. This reframes the paper's contribution from an architectural proposal to an evaluation study with a diagnosed negative result.

VMD-MFGNN (**V**ariational **M**ode **D**ecomposition **M**ulti-**F**requency **G**raph **N**eural **N**etwork):

1) Decomposes an 8-variable commodity/macro time series into $K = 5$ frequency bands via VMD, using a leakage-safe rolling window refit at every trading day;
2) Constructs a *separate* learned inter-variable graph at each frequency band, intended to capture scale-dependent cross-commodity dependencies;
3) Applies Graph Attention Networks (GAT) per frequency band for spatial modeling and an LSTM for temporal modeling;
4) Fuses band-level predictions via learned attention weights.

This paper is organized around four research questions, posed here before any results are reported so that Section V-I's answers have a clear antecedent:

- **RQ0**: Does any model tested here, including VMD-MFGNN, beat a trivial zero-forecast (do-nothing) baseline?
- **RQ1**: Does constructing a separate learned graph per frequency band outperform pooling the bands into a single graph?
- **RQ2**: Does VMD decomposition, and does end-to-end learned (versus fixed correlation-based) adjacency, add measurable value over the alternatives?
- **RQ3**: Does VMD-MFGNN differ significantly, in the Diebold-Mariano sense, from standard forecasting baselines?

Our contributions are:

- **A rigorous, leakage-safe, capacity-matched evaluation protocol**, applied to a question this literature rarely asks explicitly: does VMD decomposition and per-band graph learning add measurable forecasting value over standard baselines and a trivial naive benchmark, for copper returns at this data scale? The protocol includes a capacity-disclosed six-variant ablation (Section IV-D), a Diebold-Mariano significance test with a multiple-testing correction (Section IV), and, critically, a naive random-walk baseline comparison that this literature typically omits (Section V). Our reason for building the protocol first and treating the architecture as the object under test is the pattern Kapoor and Narayanan (2022) document across seventeen scientific fields and hundreds of papers: leakage and evaluation shortcuts, rather than modelling choices, are the dominant driver of irreproducible ML-based scientific claims, and they are typically invisible to a reader of the finished paper. This study's own history bears that out from both directions — a leaky VMD decomposer was caught and replaced before any reported result was produced (Section III), and a later re-run that appeared to reverse this paper's central negative finding turned out to be an evaluation artifact rather than a real effect (Section V-H).
- **A diagnosed, quantified mechanistic account of why the proposed graph-learning mechanism failed to produce useful structure**, not just an observation that it did: we trace the per-band adjacency's collapse to uniform edge weights to isotropic norm decay in the underlying graph embeddings during training, rule out the more obvious dead-ReLU explanation with a direct test, and identify the likely optimization dynamic responsible (Section V-E1). This is offered as a citable finding relevant to other low-node-count learned-graph architectures in financial GNN research, including MBTI-Net-style designs, not only to this paper's own model.
- **An honest reference point most comparison papers in this area omit**: every model in this study, including the proposed one, is compared directly against a trivial zero-forecast baseline, and we report plainly that none beats it on mean absolute error at any horizon (Section V).
- **A targeted, verified fix for the diagnosed collapse, and an honest account of its limits**: we engineer and test a specific intervention (unit-normalizing the graph embeddings before the bilinear product and excluding them from weight decay), confirm it mechanically resolves the norm-collapse mechanism, and show the underlying embeddings still fail to learn any task-relevant structure once the collapse itself is removed as an explanation—strengthening, not merely repeating, the earlier diagnosis (Section V-F).
- **A general methodological finding about directional accuracy**: several models across this study's experiments, including a re-run originally thought to reverse the paper's central negative result, turn out to be emitting near-constant forecasts whose apparent directional accuracy is an artifact of the test period's majority class rather than a measurement of directional skill—a failure mode invisible to a bare directional-accuracy percentage and only caught by checking prediction sign concentration and a majority-class null directly (Section V-H). We developed this check independently, but claim no priority for the underlying base-rate concern, which Cheung (2026) raises for equity forecasting; what we add is the sign-concentration and base-rate-identity machinery that detects a *degenerate forecaster*, not only a misspecified null, and can be applied retroactively to already-reported

results (Section V-H).

II. Related Work

A. Copper Price Forecasting

Traditional approaches include ARIMA (Becerra et al., 2022), GARCH (Garcia and Kristjanpoller, 2019), and VAR/VECM models (Dai, 2025). Machine learning methods—LSTM (Hu et al., 2020; Luo et al., 2022), CNN-LSTM (Li et al., 2023), XGBoost (Nabavi et al., 2024), and Transformers (Tseng and Nguyen, 2025; Zhao et al., 2025)—have shown improvements over econometric baselines, particularly at longer horizons.

B. Decomposition-Ensemble Methods

Signal decomposition followed by component-wise forecasting is the most active methodological area. VMD-LSTM (Liu et al., 2020) is, to our knowledge, the most-cited work specifically combining VMD with LSTM forecasting for non-ferrous metals, demonstrating that VMD's variational formulation avoids the mode mixing issues of EMD/EEMD (Dragomiretskiy and Zosso, 2014). Recent advances include secondary decomposition (Liu et al., 2022; Zhang et al., 2025a), adaptive VMD parameter selection (Zhang and Ke, 2025), and complexity-aware decomposition (Li and Liu, 2026). The large majority of decomposition-ensemble methods treat variables independently, discarding cross-variable information.

C. Graph Neural Networks for Financial Forecasting

MTGNN (Wu et al., 2020) introduced learned adaptive adjacency matrices for multivariate time series, and our per-band graph construction (Section III-C) follows the same adjacency parameterization applied independently within each frequency band. StemGNN (Cao et al., 2020) operates in the joint spectral-temporal domain. FourierGNN (Yi et al., 2023) unifies spatial and temporal modeling via Fourier operators. In finance, THGNN (Xiang et al., 2023) uses dynamic heterogeneous graphs for stocks, and MDGNN (Qian et al., 2024) introduces multi-relational dynamic graphs.

For commodities specifically, the literature is sparse. Sun et al. (2025) proposed a Multi-view Graph Transformer for copper that constructs a single graph over undecomposed price data—the graph structure does not vary by frequency scale. Hu et al. (2023) applied heterogeneous continual GNN to 49 Chinese futures contracts. Zhai et al. (2024) used a Graph Fourier Transformer for base metals.

D. VMD Combined with GNN, and Positioning of This Work

A small number of papers combine VMD with GNN, mostly outside finance: traffic forecasting (Ahmad et al., 2024, 2025), air quality (Pei et al., 2022), wind energy (Cheng and Tsai, 2026; Bao et al., 2025), and industrial applications (Yuan, 2025). Within finance, Li and Liu (2026) ("MBTI-Net") combines VMD decomposition with a GNN for copper futures price forecasting; this is the closest prior work to ours and rules out any claim of being first to combine VMD and GNNs for copper. Critically, MBTI-Net pools the decomposed VMD modes into a single unified graph before applying graph convolution—the graph topology does not vary across frequency bands. Zheng and Zhufu (2025) combine DCT (not VMD) with a cross-scale GNN for single-stock prediction, not multi-commodity relational modeling.

The gap we target is therefore precise: among work that combines signal decomposition with graph-based relational modeling for commodity or financial time series, we are not aware of any that constructs a *separate* graph per decomposed frequency band across a multi-commodity system, as opposed to (a) pooling bands into one graph (MBTI-Net) or (b) using a single graph over undecomposed data (Sun et al., 2025). We do not claim priority over VMD+GNN combinations generally, nor over GNN applications to copper generally. More importantly, we do not simply assert that the per-band choice is an improvement over MBTI-Net-style pooling and leave it there: we treat it as an empirical question MBTI-Net's architecture implicitly answers one way (pooling is sufficient) without testing the alternative, and we test both sides of that question directly, against a matched pooled-graph baseline (Section IV-D), at the same data scale. Our result complicates a straightforward reading of MBTI-Net's design choice: we find no ablation evidence that per-band graphs outperform pooling, and we go further than reporting that negative comparison by diagnosing *why* it comes out this way—the per-band graph-learning mechanism in our model collapsed to near-uniform edge weights during training (Section V-E1), a specific, quantified failure mode that, to our knowledge, has not previously been documented for this class of learned-adjacency GNN architecture at low node counts. It is not, however, without precedent, and we do not present it as unanticipated: Fountoulakis et al. (2023) prove, and confirm on both synthetic and real graphs, that in a low signal-to-noise ("hard") regime of the contextual stochastic block model, a graph attention network's coefficients concentrate on uniform $1/|\mathcal{N}_i|$ weights, so the attention mechanism reduces to ordinary mean-aggregating graph convolution and cannot distinguish important from unimportant edges. Their setting is attention over a *given* graph in node classification, and their mechanism is the statistical indistinguishability of intra- from inter-class edges rather than an optimization dynamic; ours is a *learned* bilinear adjacency in a forecasting model, degenerating for a different, traceable reason (Section V-E1). The qualitative endpoint — graph attention collapsing to uniform weights when there is too little signal to differentiate edges — is nonetheless the same, which we take as corroborating rather than diminishing the finding. This does not establish that MBTI-Net's own pooled graph suffers the same failure—we did not have access to retrain or inspect MBTI-Net's checkpoints—but it is a concrete, mechanistic reason to be skeptical that per-band-versus-pooled is a distinction that reliably matters at this problem scale, and a more informative contribution to that specific architectural question than either an unexamined win or an unexamined loss would be.

III. Methodology

A. Problem Formulation

Given $N = 8$ multivariate time series $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ representing copper price and related commodity/macroeconomic variables, we aim to forecast the copper price return $r_{t+h} = \ln(p_{t+h}/p_t)$ at horizons $h \in \{1, 5, 10, 22\}$ trading days, using a lookback window of L timesteps.

B. Variational Mode Decomposition

VMD (Dragomiretskiy and Zosso, 2014) decomposes a signal $f(t)$ into K band-limited intrinsic mode functions $\{u_k\}_{k=1}^{K}$ by solving:

$$\min_{\{u_k\},\{\omega_k\}} \sum_{k=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \quad \text{s.t.} \quad \sum_{k=1}^{K} u_k = f \tag{1}$$

where ω_k is the center frequency of mode k. The constrained problem is solved via ADMM in the Fourier domain with penalty parameter α.

We apply VMD independently to each of the $N = 8$ input variables with $K = 5$ modes, yielding a tensor of modes $\mathbf{U} \in \mathbb{R}^{N \times K \times T}$. To prevent temporal leakage, decomposition uses a *rolling window, refit at every trading day*: at trading day t, VMD is recomputed from scratch on the window of the most recent 252 trading days ending at and including t (`signal[max(0,t-251):t+1]`), never using data beyond t, and only the final-timestep value of each resulting mode is retained as day t's feature. Because refitting happens at every single day rather than periodically, each mode's assigned series is a genuinely daily-varying signal at the same temporal resolution as the raw price data, not a value held constant across a multi-day block. Because VMD's ADMM optimization does not guarantee a consistent mode ordering across independent daily fits, each day's K modes are resorted by ascending converged center frequency (ω_k) before being stored, so mode index 0 is always the lowest-frequency band and mode $K - 1$ the highest, consistently across the full time series. An earlier version of this pipeline instead used an expanding-window decomposer that refit only every 21 trading days and carried the resulting modes forward (constant) across each 21-day block; that design was found to collapse a 60-day lookback window into as few as three distinct feature values per variable/mode and was replaced by the daily-refit rolling-window design described above before any of the results in this paper were produced. The concern this design addresses is not idiosyncratic to this study: Feng et al. (2026) independently identify temporal leakage in VMD-based forecasting as a general problem in the electricity-demand literature — noting that decomposing an entire series, or even an entire input subsequence, before forecasting grants the model access to future information — and adopt a sample-wise, rolling-window VMD scheme for the same causality-preserving reason we adopt a daily-refit one here. Decomposed modes and refit metadata (split dates, VMD parameters, the rolling-window length, and a hash of the underlying price data) are cached to disk, with cache invalidation triggered by any change to the underlying data or parameters.

C. Frequency-Specific Graph Construction

The central architectural choice in VMD-MFGNN is constructing a *separate* graph $G_k = (V, E_k)$ for each frequency band $k \in \{1, \ldots, K\}$, where nodes $V = \{1, \ldots, N\}$ represent the $N = 8$ variables and edges E_k capture band-specific inter-variable dependencies. This is directly contrasted, via ablation, against a single graph pooled across all K bands (Section IV-D).

We employ learned adaptive adjacency matrices following Wu et al. (2020), applied independently within each band. For each band k, two embedding matrices $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)} \in \mathbb{R}^{N \times d}$ are randomly initialized and jointly optimized:

$$\mathbf{A}_k = \text{softmax}\left(\text{ReLU}\left(\mathbf{E}_k^{(1)} \mathbf{E}_k^{(2)\top} \right) \right) \tag{2}$$

(the implementation applies no $1/\sqrt{d}$ scaling to the bilinear logits before ReLU/softmax; we state this explicitly because this absence turns out to matter for the collapse mechanism diagnosed in Section V-E1). Each row of $\mathbf{A}_k$ is then sparsified to its top-$k = 4$ entries (out of $N = 8$), so the realized per-band graph is not fully connected; the complexity analysis in Section III-G accounts for this sparsification. This produces K adjacency matrices, one per frequency band, each intended to capture dependencies at its own frequency scale. As an ablation-only alternative (Section IV-D), the learned adjacency can be replaced by a fixed, precomputed adjacency built from Pearson correlation between variables' band signals, thresholded at $|r| > 0.3$ (edges below this magnitude are dropped, following a conventional weak-to-moderate correlation cutoff).

D. Frequency Band Module

For each band k, a **FrequencyBandModule** processes the k-th modes of all N variables:

Graph Attention Layer. At each timestep t, the mode values $\mathbf{u}_t^{(k)} \in \mathbb{R}^N$ are projected to hidden dimension d_h and processed by a multi-head Graph Attention Network (GAT) (Velickovic et al., 2018):

$$\mathbf{h}_t^{(k)} = \text{GAT}(\mathbf{u}_t^{(k)}, \mathbf{A}_k) + \mathbf{u}_t^{(k)} \tag{3}$$

with residual connections and layer normalization, stacked over `num_gnn_layers` GAT layers.

Temporal Layer. The GAT-enhanced representations are fed to a multi-layer LSTM across the temporal dimension, applied independently per variable, to capture sequential dependencies:

$$\mathbf{z}^{(k)} = \text{LSTM}(\mathbf{h}_1^{(k)}, \ldots, \mathbf{h}_L^{(k)}) \tag{4}$$

The output for the target variable (copper, node 0) is $\mathbf{z}_{\text{Cu}}^{(k)} \in \mathbb{R}^{d_h}$.

Algorithm 1 summarizes the full forward pass just described, from raw multivariate input through per-band graph construction and GAT+LSTM processing to the fusion and prediction steps detailed in the remainder of this section, and Figure 1 renders the same forward pass as a block diagram with tensor shapes and the per-band repetition made explicit. The two are intended to be read together: every block in

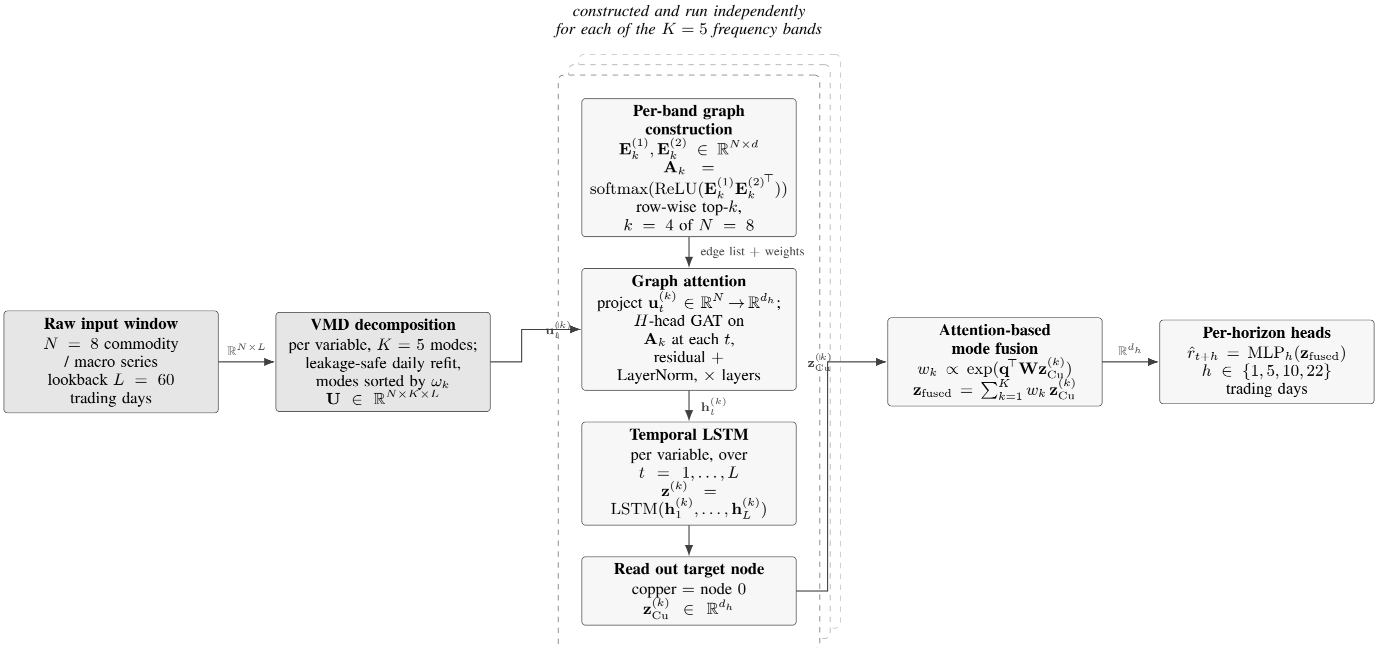

Fig. 1: VMD-MFGNN forward pass, corresponding block-for-block to Algorithm 1. An $N = 8$ variable window of length $L = 60$ trading days is decomposed by leakage-safe daily-refit VMD into $K = 5$ frequency-band modes per variable (Algorithm line 1). The dashed stack denotes the per-band branch instantiated once for each of the K bands, with its own embedding pair $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)}$ and its own adjacency $\mathbf{A}_k$, row-wise sparsified to top-$k = 4$ of $N = 8$ before being passed to the band's GAT as an edge list (lines 3–5); band-k mode values are projected to d_h and passed through the GAT at each timestep with residual connections and layer normalization, then through a per-variable LSTM over the L timesteps, and copper's representation is read from node 0 (lines 6–11). The K band representations are combined by the learned-query attention fusion of Section III (lines 13–14) and mapped to the four forecast horizons by separate heads (lines 15–16). The separate-graph-per-band structure inside the dashed stack is the architectural choice this paper tests; the pooled-graph ablation of Section IV-D replaces it with one graph shared across bands.

the figure corresponds to one or more numbered lines of the algorithm, and the figure adds no component the algorithm does not specify. It describes the architecture as designed; whether the per-band graph mechanism (lines 4–7) learns the differentiated structure it is intended to is an empirical question taken up separately in Section V-E1.

E. Attention-Based Mode Fusion

The K band-level representations are fused via a learned attention mechanism:

$$w_k = \frac{\exp(\mathbf{q}^\top \mathbf{W}\mathbf{z}_{\mathrm{Cu}}^{(k)})}{\sum_{j=1}^{K} \exp(\mathbf{q}^\top \mathbf{W}\mathbf{z}_{\mathrm{Cu}}^{(j)})}, \qquad \mathbf{z}_{\mathrm{fused}} = \sum_{k=1}^{K} w_k \cdot \mathbf{z}_{\mathrm{Cu}}^{(k)} \quad (5)$$

where $\mathbf{q}$ is a learnable query vector and $\mathbf{W}$ is a single learned linear projection (not a multi-layer network) applied to each band representation before the query dot product. The attention weights $\{w_k\}$ are intended to be interpretable as revealing which frequency bands contribute most to the prediction; Section V reports how peaked or uniform they turn out to be in practice.

F. Prediction and Training

Separate linear heads (one per horizon) map $\mathbf{z}_{\mathrm{fused}}$ to predictions: $\hat{r}_{t+h} = \mathrm{MLP}_h(\mathbf{z}_{\mathrm{fused}})$. The model is trained end-to-end with MSE loss summed across the four horizons:

$$\mathcal{L} = \sum_{h\in\mathcal{H}} \frac{1}{|\mathcal{T}|} \sum_{t\in\mathcal{T}} (r_{t+h} - \hat{r}_{t+h})^2 \quad (6)$$

The best-validation-loss checkpoint is selected and used for all reported models (Section IV-G).

G. Complexity Analysis

Let $N = 8$ be the number of variables, $T = L$ the lookback window length ($L = 60$ trading days throughout this paper, Section IV-G), $K = 5$ the number of VMD modes, d_h the hidden dimension, and H the number of GAT attention heads. Within one frequency band, the learned adjacency $\mathbf{A}_k$ is top-k sparsified to each node's $k = 4$ highest-weight outgoing edges (Section III-C) before being passed to the GAT layer, so the graph each band's GAT operates on has $O(Nk)$ edges rather than $O(N^2)$; we report the sparsified-graph cost here rather than the looser fully-connected upper bound. This gives per-timestep GAT cost $\mathcal{O}(Nk\,d_h/H)$ for the attention computation and $\mathcal{O}(Nd_h^2)$ for the linear projections, repeated over T timesteps and `num_gnn_layers` GAT layers; the per-variable LSTM that follows costs $\mathcal{O}(NTd_h^2)$. Because this cost is incurred independently for each of the K bands, VMD-MFGNN's total forward-pass cost is $\mathcal{O}(K(Nk\,Td_h + NTd_h^2))$, i.e., $K\times$ the cost of a single-graph architecture with otherwise

Algorithm 1 VMD-MFGNN forward pass

Require: Raw multivariate window $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ for $N = 8$ variables over lookback $L = 60$; number of frequency bands $K = 5$; adjacency sparsification parameter top-$k = 4$; embedding dimension d; hidden dimension d_h; number of GAT heads H; forecast horizons $\mathcal{H} = \{1, 5, 10, 22\}$
Ensure: Predicted returns $\{\hat{r}_{t+h}\}_{h \in \mathcal{H}}$
VMD decomposition: for each variable $n = 1, \ldots, N$, decompose the rolling window ending at t into K band-limited modes via leakage-safe daily-refit VMD, sort modes by ascending center frequency ω_k, and retain the final-timestep value to form $\mathbf{U} \in \mathbb{R}^{N \times K \times L}$
for $k = 1$ to K **do**
    Graph construction: initialize (or retrieve) embeddings $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)} \in \mathbb{R}^{N \times d}$
    Compute adjacency logits $\mathbf{A}_k \leftarrow \mathrm{softmax}(\mathrm{ReLU}(\mathbf{E}_k^{(1)} \mathbf{E}_k^{(2)^\top}))$
    Sparsify: retain only each row's top-$k = 4$ entries of $\mathbf{A}_k$, zeroing the rest
    Band processing: project band-k mode values $\mathbf{u}_t^{(k)} \in \mathbb{R}^N$ to hidden dimension d_h
    for $t = 1$ to L **do**
        $\mathbf{h}_t^{(k)} \leftarrow \mathrm{GAT}(\mathbf{u}_t^{(k)}, \mathbf{A}_k) + \mathbf{u}_t^{(k)}$ (residual, layer norm, stacked over `num_gnn_layers` GAT layers)
    end for
    $\mathbf{z}^{(k)} \leftarrow \mathrm{LSTM}(\mathbf{h}_1^{(k)}, \ldots, \mathbf{h}_L^{(k)})$
    Extract copper's representation $\mathbf{z}_{\mathrm{Cu}}^{(k)} \in \mathbb{R}^{d_h}$ (node 0 of $\mathbf{z}^{(k)}$)
end for
Mode fusion: for $k = 1, \ldots, K$, compute $w_k \leftarrow \exp(\mathbf{q}^\top \mathbf{W} \mathbf{z}_{\mathrm{Cu}}^{(k)}) / \sum_{j=1}^{K} \exp(\mathbf{q}^\top \mathbf{W} \mathbf{z}_{\mathrm{Cu}}^{(j)})$
$\mathbf{z}_{\mathrm{fused}} \leftarrow \sum_{k=1}^{K} w_k \cdot \mathbf{z}_{\mathrm{Cu}}^{(k)}$
for each horizon $h \in \mathcal{H}$ **do**
    $\hat{r}_{t+h} \leftarrow \mathrm{MLP}_h(\mathbf{z}_{\mathrm{fused}})$
end for
return $\{\hat{r}_{t+h}\}_{h \in \mathcal{H}}$

identical per-band capacity. This is the direct computational price of per-band graph construction, and is the reason the pooled-graph ablation (Section IV-D) is evaluated as a single-graph, single-pass alternative rather than dismissed a priori as architecturally uninteresting.

Two distinct trained instances of the VMD-MFGNN architecture appear in this paper, and their parameter counts must not be conflated. The main comparison (Section V) uses the HPO-tuned configuration selected by the search in Section IV-C ($d_h = 128$, $H = 2$, `num_gnn_layers`$= 1$), which has **1,461,124** trainable parameters, counted directly from the saved, trained model. The ablation study (Section IV-D) instead uses the untuned default configuration ($d_h = 64$, $H = 4$, `num_gnn_layers`$= 2$, per Section IV-G) for its `full_model` row, which has **393,476** trainable parameters, likewise counted from its own saved, trained checkpoint. HPO is applied only to the main-table model, never to the ablation variants (Section IV-C); consequently the ablation table's `full_model` is a genuinely different, smaller model instance than the main table's VMD-MFGNN row, not the same trained model reused across tables. This is a deliberate, disclosed design choice, not an inconsistency: re-running HPO separately for every ablation variant was judged infeasible within the project's compute budget, and holding the ablation variants at one fixed, shared default configuration keeps the ablation comparison internally consistent even though it is not directly comparable in absolute terms to the main table.

Within the ablation study, capacity is controlled two ways for the no-VMD and pooled-graph variants, and the resulting real parameter counts (from the trained checkpoints) are reported here rather than estimated: a *matched-dimension* variant that keeps the full model's d_h fixed and lets parameter count fall out of the architecture (`pooled_graph_matched_dim`: 84,618 params, -78.5% vs. `full_model`; `no_vmd_raw_price_matched_dim`: 88,836 params, -77.4%), and a *matched-parameter* variant that increases d_h until total parameter count is approximately restored to `full_model`'s (`pooled_graph_matched_params`: 397,586 params, $+1.05\%$; `no_vmd_raw_price_matched_params`: 394,116 params, $+0.16\%$). The matched-parameter variants land within about 1% of `full_model`'s parameter count, confirming the capacity-matching procedure worked as intended at real data scale. `correlation_graph` (392,196 params, -0.33%), which keeps the full per-band architecture and only replaces the learned adjacency with a fixed correlation-threshold graph, is essentially parameter-matched to `full_model` by construction (it omits only the small per-band adjacency-embedding matrices).

IV. Experimental Setup

A. Data

We use daily data from January 2010 to December 2025, sourced entirely from Yahoo Finance (no FRED, BDI, TC/RC, or COT data are used). The target variable is COMEX copper futures (HG=F). Input variables ($N = 8$) are: copper (HG=F), aluminum (ALI=F), gold (GC=F), crude oil / WTI (CL=F), US Dollar Index (DX-Y.NYB), S&P 500 (^GSPC), VIX (^VIX), and the US 10-year Treasury yield (^TNX). Zinc and nickel were originally in scope, sourced via NASDAQ commodity sub-index proxies (^NQCIZNER for zinc, ^NQCINIER for nickel) since neither has a standalone Yahoo Finance futures contract (both are LME-only instruments). On the live production data pull for this study's date range, both proxy tickers were confirmed dead/delisted (`yfinance` raised `YFPricesMissingError`), with no further viable Yahoo Finance substitute identified. Zinc and nickel were therefore removed from the locked scope entirely, rather than replaced with a different proxy—a deliberate, disclosed reduction from an originally planned 10-variable scope to the 8-variable scope reported throughout this paper, discussed further in Section VI.

Split. A single chronological train/validation/test split is used: training (2010–2019), validation (2020–2021, includes COVID), test (2022–2025, includes supply shocks, rate hikes, and green-transition demand effects). This is *not* a walk-forward or rolling-origin evaluation with multiple folds; the implications for statistical testing are discussed in Section VI.

B. Baselines

We compare VMD-MFGNN against six baselines: ARIMA, XGBoost, LSTM, Transformer, VMD-LSTM, and SimpleMT-GNN, spanning econometric, machine-learning, deep-learning, decomposition-ensemble, and graph-based categories. All six run with standard, reasonable, hand-set configurations; none receive hyperparameter optimization (Section IV-C).

C. Hyperparameter Optimization

VMD-MFGNN, the proposed model, is tuned via a lightweight Optuna study (TPE sampler with a median pruner), searching hidden dimension $\in \{32, 64, 128\}$, number of attention heads $\in \{2, 4, 8\}$ (constrained so hidden dimension is divisible by the head count), learning rate (log-uniform, 10^{-4} to 10^{-2}), dropout (uniform, 0.05 to 0.3), and number of GNN layers $\in \{1, 2, 3\}$, configured for up to 15 trials of 25 epochs each. In the run reported in this paper, the study completed 10 of the configured 15 trials (5 `COMPLETE`, 5 `PRUNED` by the median pruner before completing their allotted epochs) before the search budget was exhausted; we report the true trial count rather than the configured one. The best trial (validation MSE = 0.002051) selected **hidden dimension** $= 128$**, attention heads** $= 2$**, learning rate** ≈ 0.00153**, dropout** ≈ 0.0616**, GNN layers** $= 1$. This configuration is used for VMD-MFGNN throughout the main results (Table I) and significance testing (Table II); it is *not* used in the ablation study (Section IV-D), which fixes all variants, including its own `full_model` row, at the untuned default configuration (Section IV-G) instead. The six baselines are *not* tuned; they run at standard/hardcoded settings. This asymmetry is a deliberate, disclosed methodological choice rather than an oversight: tuning all seven models under a full hyperparameter search would multiply compute cost roughly by the trial count and was judged infeasible within this project's compute and time budget, and tuning only the proposed model against standard baselines is common, defensible practice in comparison papers under such constraints, provided it is disclosed (Section VI). HPO results (best parameters and the full trial history, including pruned trials) are saved for reproducibility.

D. Ablation Studies

Six variants isolate the contribution of each architectural component, all trained at the untuned default configuration (Section IV-G, Section III-G)—HPO (Section IV-C) is applied only to the main-table VMD-MFGNN, never to any ablation variant. Because this study uses a single chronological split rather than multiple walk-forward folds or multiple random seeds (Section VI), the ablation results below are reported as point estimates per horizon, without a paired significance test; a valid resampling unit (folds or seeds) does not exist for *this* run to compute one. A separate five-seed repetition of the two variants that carry RQ1, run later at the tuned configuration and therefore not directly comparable to this table, does admit a paired test and is reported in Section V-G.

- **(a) Full model**: VMD decomposition into $K = 5$ bands, with a separate learned graph per band, at the untuned default configuration (393,476 parameters). This is a distinct trained instance from the HPO-tuned VMD-MFGNN reported in Table I (1,461,124 parameters); see Section III-G.
- **(b) No-VMD, raw price**: A genuine raw-price GAT+LSTM model with a single learned graph, using a dedicated raw-price dataloader that never touches VMD-decomposed input, reported at two capacity settings: *matched-dim* (same hidden dimension as the full model, 88,836 params, -77.4%) and *matched-params* (hidden dimension increased to restore parameter count, 394,116 params, $+0.16\%$). This isolates the contribution of decomposition itself, independent of the per-band-graph question, at two different capacity assumptions.
- **(c) Pooled graph**: VMD decomposition retained, but the K band representations are pooled into a single unified representation before a single graph is built over it, rather than building K separate graphs. This is the ablation that most directly supports (or fails to support) the paper's central architectural claim against MBTI-Net-style pooling. Reported at two capacity settings: *matched-dim* (84,618 params, -78.5%) and *matched-params* (397,586 params, $+1.05\%$). The matched-params setting lands within about 1% of the full model's parameter count, so the (a)-vs-(c, matched-params) comparison is the cleanest available test of graph structure with capacity approximately controlled; the matched-dim setting is retained alongside it because it reflects the capacity a pooled architecture has *by default* at equal hidden dimension, which is itself an architecturally meaningful comparison.
- **(d) Correlation graph**: The learned per-band adjacency is replaced with a fixed adjacency built from Pearson correlation between variables' band signals, thresholded at $|r| > 0.3$ (392,196 params, -0.33%, essentially parameter-matched to the full model). This isolates the contribution of end-to-end learned adjacency versus a fixed, precomputed graph, holding the per-band architecture fixed.

E. Statistical Testing

For the baseline comparison (VMD-MFGNN versus each of the six baselines, Section IV-B), we apply the Diebold-Mariano test (Diebold and Mariano, 1995) to each pair of forecast error series, per horizon, testing whether VMD-MFGNN's forecast errors are significantly different from each baseline's. This test is applicable here because the same test-set forecasts, produced under the same single split, provide a valid paired sample of loss differentials over time. The test statistic is computed on the loss differential $d = e^2_{\text{VMD-MFGNN}} - e^2_{\text{baseline}}$, divided by the square root of a heteroskedasticity-and-autocorrelation-consistent (HAC) variance estimate that sums d's sample autocovariances at lags $1, \ldots, \text{horizon} - 1$ with equal (rectangular-kernel) weight; this is a simpler, unweighted HAC-style correction, not the Bartlett-weighted kernel of the classical Newey-West estimator, and we describe it as such here to match the implementation exactly rather than name it Newey-West. Consequently a *negative* DM statistic indicates

VMD-MFGNN's squared error was lower (VMD-MFGNN better) and a *positive* DM statistic indicates the baseline's squared error was lower (baseline better), and this sign convention is used without repeating it per cell in Table II. Because 24 DM tests are run (six baselines $\times$ four horizons), we additionally apply the Holm-Bonferroni step-down procedure to control the family-wise error rate across all 24 tests at $\alpha = 0.05$, and report both the raw and Holm-corrected significance in Table II (Section V-B). The DM test is *not* applied to the ablation comparisons (Section IV-D): with only one train/validation/test split and no walk-forward folds or repeated seeds in that run, there is no resampling unit across which to compute a paired test or an effect size such as Cohen's d for the ablation variants as reported in Table III, and we do not fabricate one. Where seeds *are* available — the five-seed repetition of the two RQ1 variants at the tuned configuration, Section V-G — we do report paired t- and Wilcoxon signed-rank tests over seeds, but that is a separate experiment at a different configuration and its tests do not transfer to Table III.

F. Evaluation Metrics

RMSE, MAE, and Directional Accuracy (DA) are reported for the main comparison and the ablations, per horizon.

G. Implementation Details

VMD-MFGNN and all baselines are implemented in PyTorch with PyTorch Geometric for the GAT layers. Default (pre-HPO) architecture hyperparameters are: hidden dimension 64, 4 attention heads, 2 GNN layers, 2 LSTM layers, dropout 0.1, lookback window $L = 60$ trading days; training uses batch size 32, the Adam optimizer with learning rate 10^{-3} and coupled L_2 weight decay 10^{-5} applied uniformly to all parameters (including the graph-embedding parameters discussed in Section V-E1), and early stopping. Best-validation-loss checkpoint selection was used for all reported models, with model weights saved to disk so that training can resume from a checkpoint rather than restart from scratch.

V. Results and Discussion

All numbers below are computed on the test split (2022–2025) and read directly from the training and evaluation pipeline's saved output artifacts, with no recomputation or adjustment beyond the naive-baseline row, the Holm-Bonferroni correction, and the ablation-table cross-checks described explicitly where they appear.

A. Main Results, Including a Naive Baseline

Table I reports RMSE, MAE, and Directional Accuracy (DA) for VMD-MFGNN and the six baselines, across all four forecast horizons, together with a trivial **zero-forecast (random-walk) baseline** that predicts $\hat{r}_{t+h} = 0$ for every sample. We add this row because it is the single most important context for reading the rest of this table, and because omitting it, as much of the decomposition-ensemble and GNN-for-finance literature does, risks presenting a ranking among models that all fail a more basic test. For a zero forecast, RMSE reduces to the standard deviation of the true returns and MAE to their mean absolute value, computed directly from the saved test-set target arrays (identical across all seven models since they share one test set).

TABLE I: Forecasting performance comparison across four horizons, test split (2022–2025). Best result per column shown in **bold**. VMD-MFGNN is the HPO-tuned model (Section IV-C); the six baselines are untuned. ‡Transformer's $h = 5$ DA is a confirmed base-rate artifact: its predictions are 100% positive-sign and this DA value exactly equals the test period's up-day frequency, not a measurement of directional skill; see the sign-concentration diagnostic in Section V-H.

Model	RMSE				MAE				DA (%)			
	h=1	h=5	h=10	h=22	h=1	h=5	h=10	h=22	h=1	h=5	h=10	h=22
ARIMA	0.0182	0.0403	0.0555	0.0760	0.0122	0.0281	0.0399	0.0572	49.4	48.7	49.3	**49.9**
XGBoost	0.0180	0.0399	**0.0551**	0.0765	0.0121	0.0279	0.0397	0.0578	46.8	51.2	48.8	42.7
LSTM	0.0193	0.0416	0.0606	0.0834	0.0136	0.0288	0.0468	0.0675	48.7	**53.7**	45.2	45.9
Transformer	0.0231	0.0493	0.0575	0.1053	0.0177	0.0357	0.0420	0.0870	51.9	53.0‡	**52.7**	44.7
VMD-LSTM	0.0453	0.0458	0.0637	0.0961	0.0366	0.0334	0.0474	0.0765	48.9	49.5	51.5	48.9
SimpleMTGNN	0.0187	0.1345	0.0667	0.0977	0.0130	0.1087	0.0537	0.0782	48.3	47.3	44.5	47.8
VMD-MFGNN	0.0184	**0.0398**	0.0552	0.0785	0.0125	0.0282	0.0407	0.0608	**52.5**	50.8	47.7	47.1
Zero-forecast (naive)	**0.0179**	0.0399	0.0552	**0.0759**	**0.0120**	**0.0279**	**0.0397**	**0.0571**	—	—	—	—
Unconditional up-day freq. (%)									51.0	53.0	55.4	55.3

Table I makes the central finding of this paper visible in one place: **the naive zero-forecast baseline wins MAE at all four horizons and RMSE at two of four horizons ($h = 1$ and $h = 22$), against every one of the seven fitted models, including VMD-MFGNN.** No model beats it on MAE anywhere in the table; VMD-MFGNN and XGBoost each beat it on RMSE at one horizon ($h = 5$ and $h = 10$ respectively), and nowhere else does any model do better than predicting zero return. All seven models' R^2 values are negative or approximately zero on this test split (not tabulated here for space, computed identically to RMSE/MAE from the same saved arrays), consistent with none of them explaining variance beyond what a constant forecast already captures. DA tells a related story once read against the correct null: the relevant baseline for DA is not 50% but the test period's own unconditional up-day frequency, which is 51.0%, 53.0%, 55.4%, and 55.3% at $h = 1, 5, 10, 22$ respectively (bottom row) — itself a symptom of the strong upward drift copper exhibited over 2022–2025. The complementary down-day frequencies are 48.37%, 46.95%, 44.21%, and 44.72% (they do not sum to exactly 100% with the up-day figures because a small number of test days record an exactly zero return); we state them explicitly because the base-rate-identity checks reported below and in Section V-C are made against whichever of the two frequencies matches the sign a degenerate model emits, and a reader cannot otherwise verify those flags from the table alone. VMD-MFGNN's headline $h = 1$ DA of 52.5% is only 1.5 percentage points above that null, and at $h = 5, 10, 22$ no model in this table, including VMD-MFGNN, exceeds the unconditional up-day frequency at all. We read this table as showing that this study's seven fitted models, including the proposed one, are best understood as a ranking among approaches that do not clearly outperform doing nothing, rather than as a demonstration that any of them, including VMD-MFGNN, has found exploitable structure in copper returns at this data scale. Section V-B still reports pairwise significance among the seven fitted models, since that question — do these models differ meaningfully from one another — is separate from, and does not resolve, the more basic question this section answers in the negative. We applied this paper's own base-rate-identity check (Section V-H) to every cell of this table directly, loading all 28 saved prediction arrays and checking both the fraction of predictions sharing the majority sign and whether DA exactly equals the test period's class frequency: exactly one cell qualifies, Transformer at $h = 5$ (100% positive-sign predictions, DA=53.05% identical to the up-day frequency to four decimal places, marked ‡ above). Transformer's $h = 22$ DA (44.72%) also happens to numerically equal the down-day frequency at that horizon, but this is a coincidental count match, not the same artifact: only 89.4% of its predictions are negative-sign and its prediction standard deviation is a healthy fraction of the true-return standard deviation, so we do not flag it. No other cell in this table shows either signature.

Figure 2 shows the seven-model comparison graphically (grouped bar charts of RMSE, MAE, and DA by model and horizon; the naive baseline is not part of this bar chart, which mirrors the pipeline's model-comparison output). No model's saved evaluation record carried an error flag, so all seven models are represented in both Table I and Figure 2.

B. Statistical Significance versus Baselines

Table II reports the Diebold-Mariano statistic and p-value for VMD-MFGNN's forecast errors against each of the six baselines, per horizon (Section IV). Recall the sign convention (Section IV): the DM statistic is computed as $e^2_{\text{VMD-MFGNN}} - e^2_{\text{baseline}}$, so a *negative* DM statistic means VMD-MFGNN had the lower squared error (VMD-MFGNN better), and a *positive* DM statistic means the baseline had the lower squared error (baseline better); boldface marks $p < 0.05$. This test is applied only to the baseline comparison, not to the ablation table (Table III); see Section VI. All 24 test entries are finite (no degenerate $\text{var}_d \leq 0$ cases were observed at full data scale).

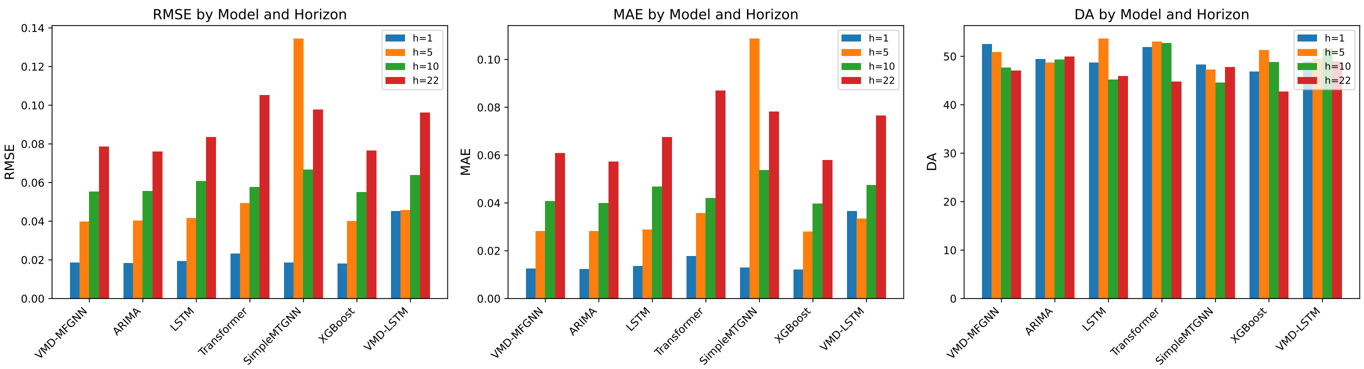

Fig. 2: RMSE, MAE, and Directional Accuracy by model and horizon, for VMD-MFGNN and the six baselines.

TABLE II: Diebold-Mariano test: VMD-MFGNN versus each baseline, per horizon. Negative DM statistic favors VMD-MFGNN; positive favors the baseline. Bold p-values indicate significance at $\alpha = 0.05$ uncorrected; † marks the subset that remains significant after Holm-Bonferroni correction across all 24 tests.

	DM statistic				p-value			
vs. Baseline	h=1	h=5	h=10	h=22	h=1	h=5	h=10	h=22
ARIMA	1.111	−0.644	−0.171	0.867	0.267	0.520	0.864	0.386
XGBoost	2.459	−0.150	0.092	0.781	**0.014**	0.880	0.927	0.435
LSTM	−2.522	−1.012	−3.595	−1.489	**0.012**	0.311	**0.0003**†	0.137
Transformer	−11.219	−3.319	−1.101	−3.734	**<0.001**†	**0.001**†	0.271	**<0.001**†
VMD-LSTM	−16.777	−3.464	−2.696	−2.217	**<0.001**†	**0.001**†	**0.007**	**0.027**
SimpleMTGNN	−0.860	−7.872	−4.050	−2.304	0.390	**<0.001**†	**<0.001**†	**0.021**

At the uncorrected $\alpha = 0.05$ level, VMD-MFGNN attains a statistically significant advantage (negative DM statistic, $p < 0.05$) over VMD-LSTM at all four horizons, over Transformer at three of four horizons ($h = 1, 5, 22$), over SimpleMTGNN at three of four horizons ($h = 5, 10, 22$), and over LSTM at two of four horizons ($h = 1, 10$; two of four is not a majority, and we state it as two of four rather than describing it loosely as most horizons)—12 of 24 comparisons favor VMD-MFGNN significantly, plus one significant result in the opposite direction (XGBoost at $h = 1$, below), for 13 of 24 raw-significant cells in total. Because 24 tests were run without correction, we also apply the Holm-Bonferroni step-down procedure at family-wise $\alpha = 0.05$ (Section IV); this reduces the significant count from 12 (favoring VMD-MFGNN) to 8, marked † in Table II, and, notably, removes VMD-MFGNN's only significant loss: XGBoost's $h = 1$ advantage ($p = 0.014$ raw) does not survive correction (Holm threshold at that step is 0.00357). After correction, VMD-MFGNN's remaining significant advantages are concentrated at shorter-to-medium horizons against Transformer ($h = 1, 5, 22$), VMD-LSTM ($h = 1, 5$), SimpleMTGNN ($h = 5, 10$), and LSTM ($h = 10$ only). Against ARIMA, no horizon reaches significance in either direction, corrected or not. Against XGBoost, no horizon is significant after correction; we still report the raw, uncorrected $h = 1$ result ($p = 0.014$, XGBoost lower squared error) in the table for transparency, but do not carry it forward as a supported claim once multiple testing is accounted for. VMD-MFGNN's significant edge in this study, corrected or uncorrected, is concentrated against the deep-learning baselines (LSTM, Transformer, VMD-LSTM, SimpleMTGNN), not against the two simpler, non-deep-learning baselines (ARIMA, XGBoost). Section V-A's naive-baseline comparison is the more important context for interpreting any of these significant differences: a significant DM result establishes that two models' squared errors differ, not that either model is exploiting real structure in copper returns, which Table I indicates neither is doing relative to a zero forecast.

C. Ablation Study

Table III reports RMSE, MAE, and DA per horizon for the six ablation variants defined in Section IV-D, all trained at the untuned default configuration. No significance column is included: with a single chronological split and no walk-forward folds or repeated seeds, there is no resampling unit available to compute a paired test or effect size for these comparisons (Section VI). Reported parameter counts are exact, read from the trained checkpoints (Section III-G), not estimated.

Figure 3 shows the same comparison graphically.

This table does not support the paper's central architectural claim, and we report that plainly, stated precisely rather than more strongly than the numbers allow. The precise claim is: row (a), the per-band full model, never attains the *best-of-six* value in any of the twelve RMSE/MAE/DA columns. This is a different and weaker claim than saying the full model loses every head-to-head comparison against either pooled variant specifically, and it does not — the full model in fact wins several individual head-to-head columns against both pooled variants, just never

TABLE III: Ablation study results, all four horizons, test split. All variants trained at the untuned default configuration (Section IV-G); this is a different, smaller VMD-MFGNN instance than the HPO-tuned model in Table I. No significance column (single-split protocol; see Section VI). Best result per column shown in **bold**. ‡Marks DA values confirmed as base-rate artifacts: each equals the test period's class frequency for that horizon exactly (Section V-H); two of these (‡ cells in the matched-dim $h = 5$ and matched-params $h = 22$ columns) would otherwise be the bolded best-in-column value and are left unbolded for this reason.

	RMSE				MAE				DA (%)			
Variant	h=1	h=5	h=10	h=22	h=1	h=5	h=10	h=22	h=1	h=5	h=10	h=22
(a) Full model (393,476 params)	0.0183	0.0403	0.0563	0.0787	0.0125	0.0290	0.0420	0.0617	48.6	49.7	46.4	48.1
(b) No-VMD, matched-dim (88,836)	0.0279	0.0400	0.0569	0.0763	0.0231	0.0280	0.0429	0.0575	48.4‡	53.0‡	46.1	49.4
(b) No-VMD, matched-params (394,116)	**0.0181**	0.0399	0.0556	0.0816	**0.0123**	0.0281	0.0404	0.0645	48.4‡	51.5	44.2‡	44.7‡
(c) Pooled graph, matched-dim (84,618)	0.0185	**0.0392**	**0.0546**	0.0788	0.0128	**0.0278**	**0.0394**	0.0610	**51.1**	51.4	**52.8**	44.6
(c) Pooled graph, matched-params (397,586)	0.0225	0.0419	0.0552	0.0762	0.0168	0.0291	0.0401	**0.0568**	50.9	52.9	49.8	55.3‡
(d) Correlation graph (392,196)	0.0183	0.0402	0.0551	**0.0754**	0.0125	0.0288	0.0406	0.0577	48.8	48.2	49.9	46.6

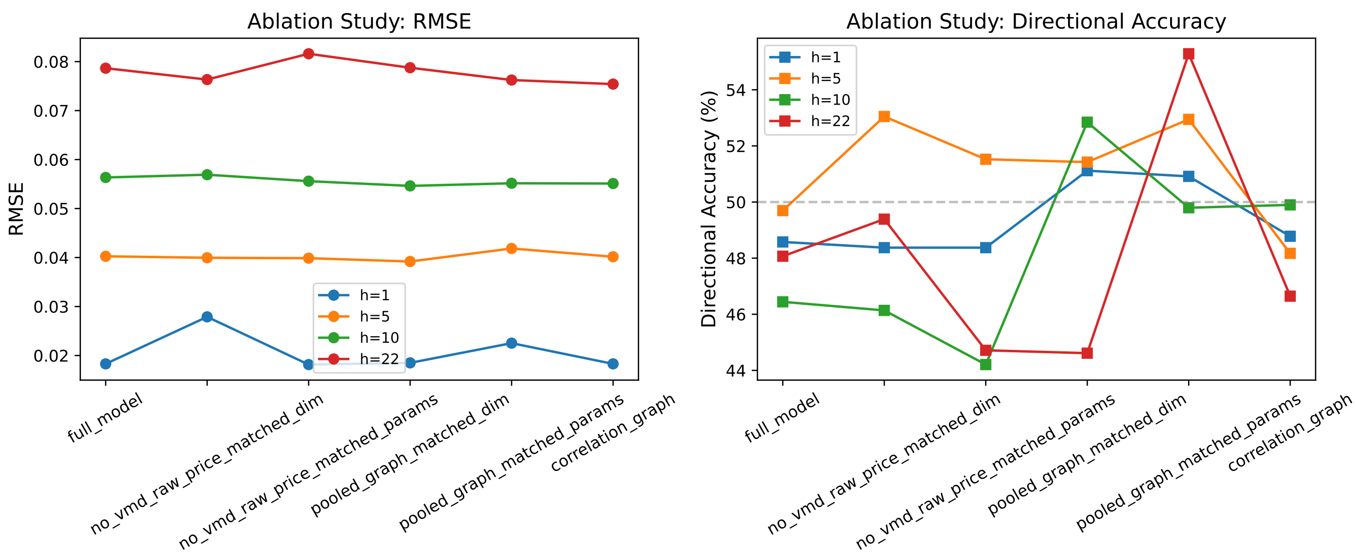

Fig. 3: RMSE and Directional Accuracy across the six ablation variants, one line per horizon.

enough to be the single best of all six variants in any column. Against `pooled_graph_matched_params` (397,586 vs. 393,476 params, +1.05%, the comparison most directly targeting RQ1 with capacity approximately controlled), the full model wins RMSE at $h = 1$ and $h = 5$ and MAE at $h = 1$ and $h = 5$; against `pooled_graph_matched_dim` (84,618 params, -78.5%), the full model wins RMSE at $h = 1$ and $h = 22$, MAE at $h = 1$, and DA at $h = 22$. What the full model does not do, against either pooled variant, is show a directional-accuracy advantage that holds across horizons: `pooled_graph_matched_params` beats the full model on DA at *three of four* horizons ($h = 1$: 48.6% vs. 50.9%; $h = 5$: 49.7% vs. 52.9%; $h = 10$: 46.4% vs. 49.8%), and `pooled_graph_matched_dim` beats it on DA at the same three horizons ($h = 1, 5, 10$). The full model's *only* directional-accuracy win over either pooled variant anywhere in this table is the $h = 22$ win against `pooled_graph_matched_dim` already counted above (48.1% vs. 44.6%), against a variant carrying 78.5% fewer parameters; it does not win DA against the capacity-matched pooled variant at any horizon where the comparison is interpretable. The nominal fourth win for `pooled_graph_matched_params`, at $h = 22$ (55.3% vs. 48.1%), is not counted here: as flagged in Table III and explained below, that cell is a confirmed base-rate artifact, so the $h = 22$ per-band-versus-capacity-matched-pooled DA comparison yields no interpretable result in either direction. Taken together: the per-band full model is competitive with, and sometimes better than, the pooled-graph variants on RMSE/MAE, particularly at short horizons, but the single directional-accuracy win it does record — one horizon, against the -78.5%-capacity pooled variant only — does not amount to the per-band advantage the architecture was designed to produce, and it never wins outright against the full six-way comparison on any metric. This is the honest summary we take forward into Section V-I: not that per-band graphs lose everything, but that they do not demonstrate the advantage the architecture was designed to produce. We flag one caveat on the DA columns specifically, in light of a finding we only made later while investigating a subsequent experiment (Section V-H): some models in this study's ablations turn out to be emitting near-constant, near-zero-variance forecasts whose directional accuracy is an artifact of the test period's class frequencies rather than a measurement of real directional skill. We did not archive per-sample prediction arrays for this untuned ablation run, so we cannot directly check prediction dispersion or sign concentration for these variants; but the base-rate-identity half of the check (Section V-H) requires only a variant's reported DA value and the test period's class frequency at that horizon, both of which are already in hand from Table I and Table III, and we apply it directly to every cell of this table. Six cells match a class frequency to four decimal places: `no_vmd_raw_price_matched_dim` at $h = 1$ (48.37%) and $h = 5$ (53.05%), `no_vmd_raw_price_matched_params` at $h = 1$ (48.37%), $h = 10$ (44.21%), and $h = 22$ (44.72%), and `pooled_graph_matched_params` at $h = 22$ (55.28%); these are marked ‡ in Table III and are not treated as demonstrating real directional skill anywhere in this section's conclusions. The remaining DA cells, and all RMSE/MAE cells, do not match a class frequency and are read at face value.

The no-VMD ablation (row (b)) is similarly mixed rather than a clean win for decomposition: the matched-params variant, which omits VMD entirely and consumes raw prices with a single per-band-shaped graph, attains the best RMSE and MAE of *all six variants* at $h = 1$, and is the worst performer on RMSE of all six variants at $h = 22$ (0.0816). On DA at $h = 22$, however, the no-VMD matched-params variant (44.7%) is *not* the worst of the six — `pooled_graph_matched_dim` is lower still (44.6%) — so "worst on RMSE and DA at $h = 22$" is not accurate for this variant; only the RMSE half of that claim holds. The matched-dim no-VMD variant is markedly worse than every other variant on RMSE/MAE at $h = 1$ (a raw, undecomposed price series appears harder for a small-capacity model to fit directly), and recovers to be competitive on RMSE/MAE at $h = 5$; its nominally best-of-six DA at $h = 5$ (53.0%) is one of the six confirmed base-rate artifacts identified above (Section V-H), so we do not read this variant as demonstrating a real directional-accuracy win at that horizon. The correlation-graph variant (row (d)), which keeps VMD and the per-band architecture but replaces the learned adjacency with a fixed correlation-threshold graph, wins the best RMSE of all six variants at $h = 22$ and is close to the full model everywhere else—so even the comparison isolating learned versus fixed adjacency does not show a clear advantage for end-to-end graph learning over the untuned full model at this configuration.

We emphasize that these ablation results describe the untuned default-configuration model only (Section IV-D); they are not a statement about the HPO-tuned VMD-MFGNN reported in Table I, which is a substantially larger, differently configured model instance. Section V-I discusses what these two, not-directly-comparable result sets jointly imply for RQ1–RQ3.

D. Confound-Check at Tuned Hyperparameters

Table III's `full_model` and `pooled_graph_matched_params` rows are both trained at the untuned default configuration (Section III-G), while the main-table VMD-MFGNN (Table I) is separately tuned; a fair per-band-versus-pooled comparison therefore requires re-running both variants at the same, HPO-tuned configuration, since architecture and hyperparameter budget were confounded in the original ablation. We re-trained `full_model` and `pooled_graph_matched_params` at the tuned configuration selected in Section IV-C (hidden dimension 128, 2 attention heads, dropout 0.0616, 1 GNN layer), saved separately as `full_model_tuned` and `pooled_graph_matched_params_tuned` (`results/confound_check_tuned.json`), so as not to overwrite the original, already-reported ablation run.

At the tuned configuration (Table IV), the directional-accuracy picture visibly reverses relative to the untuned ablation: `full_model_tuned` now beats `pooled_graph_matched_params_tuned` on DA at three of four horizons ($h = 1$: 52.24 vs. 48.37; $h = 5$: 53.25 vs. 53.05; $h = 22$: 49.80 vs. 44.72), losing clearly only at $h = 10$ (49.29 vs. 55.39); mean squared error is essentially tied, with a small edge to the pooled variant (avg. MSE 0.002808 vs. 0.002686). Read on its own, this looks like exactly the confound-resolution result the untuned ablation's Limitations disclosure (Section VI) called for: once hyperparameters are no longer confounded with architecture, per-band graph construction appears to help directional accuracy after all. We checked the underlying adjacency directly before drawing that conclusion, the same way we checked it for the main HPO-tuned model in Section V-E1, and found the same collapse signature in `full_model_tuned`'s checkpoint: embedding RMS at 2.9–3.3% of Xavier-initialization magnitude across all five bands, softmax standard deviation on the order of 2×10^{-5}—indistinguishable from the collapse already documented for the main model. Whatever is producing `full_model_tuned`'s DA advantage in this table, it is not a graph that learned real relational structure, since that graph is mathematically uniform to the same degree as every other checkpoint inspected in this paper. We do not, however, take this table's DA columns at face value as evidence of an architecture-alone advantage either, for a reason that only became clear once we investigated the pooled variant's predictions directly: Section V-H shows `pooled_graph_matched_params_tuned`'s DA values in this very table are, to four decimal places, exactly the test period's unconditional class frequency at each horizon (48.3740, 53.0488, 55.3862, 44.7154, matching 48.37, 53.05, 55.39, 44.72 above), the signature of a degenerate, near-constant-sign forecaster rather than a model exercising directional skill. We therefore withdraw any reading of this table as demonstrating a real per-band directional-accuracy advantage: the comparison is contaminated by (at least) one side reporting an artifact, and Section V-H explains why, and how we caught it, in full.

E. Interpretability Analysis

1) Frequency-Specific Inter-Commodity Graphs: Figure 4 visualizes the learned adjacency matrices $\mathbf{A}_k$ for each of the $K = 5$ frequency bands, read from the HPO-tuned

TABLE IV: Confound-check: full model versus pooled-graph, both retrained at the HPO-tuned configuration (Section IV-C), test split. Compare against Table III's untuned rows for the same two variants.

	RMSE				MAE				DA (%)			
Variant	h=1	h=5	h=10	h=22	h=1	h=5	h=10	h=22	h=1	h=5	h=10	h=22
Full model (tuned)	0.0181	0.0397	0.0558	0.0788	0.0124	0.0279	0.0407	0.0595	52.24	53.25	49.29	49.80
Pooled graph, matched-params (tuned)	0.0185	0.0400	0.0551	0.0759	0.0127	0.0279	0.0393	0.0572	48.37	53.05	55.39	44.72

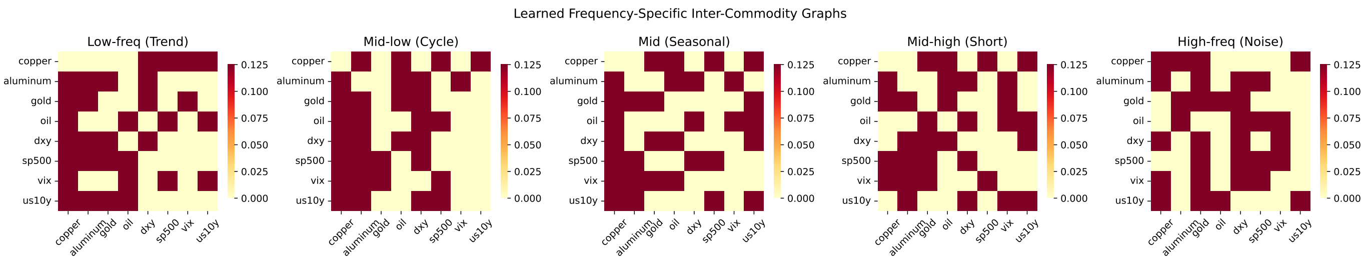

Fig. 4: Learned adjacency matrices for each of the five VMD frequency bands.

VMD-MFGNN's trained `FrequencyGraphConstructor` output.

Every kept edge weight in every one of the five bands is $\approx 0.125 = 1/8$ (the reciprocal of the $N = 8$ node count) to five decimal places. This is not a subtle near-uniformity that a more sensitive statistic might resolve into graded structure; it is uniformity as precise as floating-point softmax output gets. An earlier verification pass over this same model had reported "genuinely differentiated" per-band graphs, based on the fact that *which* edges survive top-k sparsification differs somewhat across bands. That finding was real but shallow: it checked which edges the top-k operator selected, not the values of the weights themselves, and those weights are uniform. The apparent per-band differentiation in edge *selection* turns out to come from residual, fifth-decimal-place noise inherited from each band's independent random initialization, not from learned, task-relevant structure — selecting top-k among near-identical values is highly sensitive to exactly this kind of noise. We therefore withdraw the earlier framing rather than soften it: the learned per-band adjacency mechanism did not produce differentiated relational structure of any kind, topological or graded, in the trained model this paper reports.

We diagnosed the mechanism behind this collapse directly from the trained model's checkpoints (no retraining required), and report the mechanism rather than only the symptom, since it is the more generalizable and more citable of the two. The most obvious candidate explanation — that ReLU has "died" and is clipping all pre-softmax logits to zero, which would trivially produce a uniform softmax output — does not hold up under a direct test: the fraction of negative pre-activation values is statistically indistinguishable before and after training (50.5% at Xavier initialization vs. 56.3% post-training), and removing the ReLU nonlinearity from the adjacency computation entirely still yields a uniform adjacency to four decimal places. The real mechanism is **isotropic norm collapse** of the graph embeddings $\mathbf{E}_k^{(1)}, \mathbf{E}_k^{(2)}$ (Section III-C): RMS embedding magnitude fell from its Xavier-initialization value of 0.2885 to 0.0089 over training, a factor of approximately 32, which drives the bilinear adjacency logits ($\mathbf{E}_k^{(1)}\mathbf{E}_k^{(2)\top}$) from a standard deviation of 0.331 at initialization to 2.8×10^{-4} post-training — small enough to place softmax in its first-order, near-linear regime, where it returns approximately $1/N$ regardless of the input logits' relative ordering. Critically, no directional learning occurred alongside this shrinkage: two independently trained models (differing in hidden dimension, learning rate, and training duration — the HPO-tuned and untuned default configurations of Section III-G) retained embedding directions with cosine similarity exceeding 0.99, differing essentially only by an overall scale factor; rescaling either model's trained embeddings back to their initialization norm reproduces a softmax spread statistically indistinguishable from a fresh random draw. In other words, training rescaled these embeddings without ever rotating them into a task-informed direction — the surviving direction encodes no more structure than the random initialization did. We attribute this to the interaction between coupled L_2 weight decay (Adam with weight_decay= 10^{-5}, applied uniformly to all parameters including these embeddings, Section IV-G), which exerts a constant, sign-consistent force pulling every embedding parameter toward the origin, and a task gradient reaching these specific parameters that is weak by comparison (measured on the order of 10^{-5} per element) because a near-uniform adjacency is already a close-to-optimal mean-aggregation operator for this architecture, whose LSTM temporal pathway carries most of the model's predictive signal — leaving little loss pressure to differentiate edges in the first place. This makes the embedding norm a self-reinforcing attractor toward zero: the task gradient shrinks roughly in proportion to $\|\mathbf{E}\|$ while the decay force stays constant relative to $\|\mathbf{E}\|$, so the ratio between them degrades monotonically once collapse begins, with no mechanism to recover. This finding is independent of model capacity — it is observed with near-identical severity at both the default (hidden dimension 64) and HPO-tuned (hidden dimension 128) configurations — and instead tracks training duration (RMS-to-init ratio: 0.92 at epoch 20, 0.47 at epoch 25, 0.031 at epoch 32, 0.012 at epoch 43), consistent with a genuine optimization dynamic rather than an architectural capacity limitation.

Figure 5 makes the two halves of this mechanism visible side by side, using the 19-cell robustness suite of Section V-G rather than the single model discussed above, because that suite is the only place in this study where embedding geometry was measured across many independently trained checkpoints. Panel (a) is the collapse itself: relative embedding magnitude falls across six and a half orders of magnitude, from 33% of initialization at five epochs of training to 9×10^{-8} of it at thirty-seven, and the descent is orderly rather than configuration-dependent. Panel (b) is the half that makes the finding a diagnosis rather than an observation: over that same span the embeddings' cosine similarity to their own random initialization moves by a single factor of 2.2, from 0.93 to 0.42, and does so as a smooth function of training duration alone. A mechanism that had found task-relevant structure would show the opposite disproportion — direction moving decisively while magnitude stayed bounded. What the two panels show instead is embeddings shrinking essentially in place.

Two clarifications about what this failure mode is and is not. It is *not* over-smoothing, nor representational rank collapse in the sense of Roth and Liebig (2024): those are depth-driven phenomena in which node *representations* progressively lose rank as message passing iterates, governed by the spectrum of the aggregation operator. The models reported here use one (HPO-tuned, Section IV-C) or two (default, Section IV-G) GAT layers, so the depth precondition for those phenomena is not met, and the collapse we document is in the adjacency-*generating* embedding parameters, upstream of any message passing, with gradients that flow correctly and are merely far too weak. Conflating the two would misattribute an optimization failure to an architectural one. It is, on the other hand, consistent with the theoretical picture of Fountoulakis et al. (2023) (Section II), who show that graph attention's coefficients concentrate on uniform weights whenever the underlying signal is too weak for *any* attention function to separate edges. Our contribution relative to that result is an empirical, mechanistically traced instance of the same degenerate endpoint reached by a different route — an optimization dynamic acting on a learned bilinear adjacency — in a financial forecasting setting at $N = 8$ nodes.

We return to what this implies for the ablation results and for learned-graph GNN architectures generally in Section V-I.

2) Mode Importance: Figure 6 shows the fusion attention weights (Section III), read from the model's saved mode-importance output. This artifact contains only 24 samples (the model's last, partial evaluation batch), not the full 984-sample test set used for every other reported figure in this paper; we state this sample count explicitly because it materially limits what can be claimed from it. Within these 24 samples, the mode-fusion attention weights are close to uniform across the $K = 5$ bands, ranging from approximately 0.19 to 0.22. We do not report a trend or drift over the test period from this artifact: 24 samples is too small a window, and too unrepresentative of the four-year test period as a whole (it reflects only the final evaluation batch), to support a claim about how attention evolves over time, and an earlier draft's description of a "gradual drift over the test period" from this same data is not defensible at this sample size and has been removed. What the near-uniform range across those 24 samples does support, consistent with the adjacency-collapse finding above, is that no single frequency band dominates fusion sharply in this trained model; we do not extend that observation further than the data allow.

Figure 7 shows the VMD decomposition itself (copper price and its five frequency-band modes), included here for completeness as the input representation that both the per-band graphs and the fusion attention operate on.

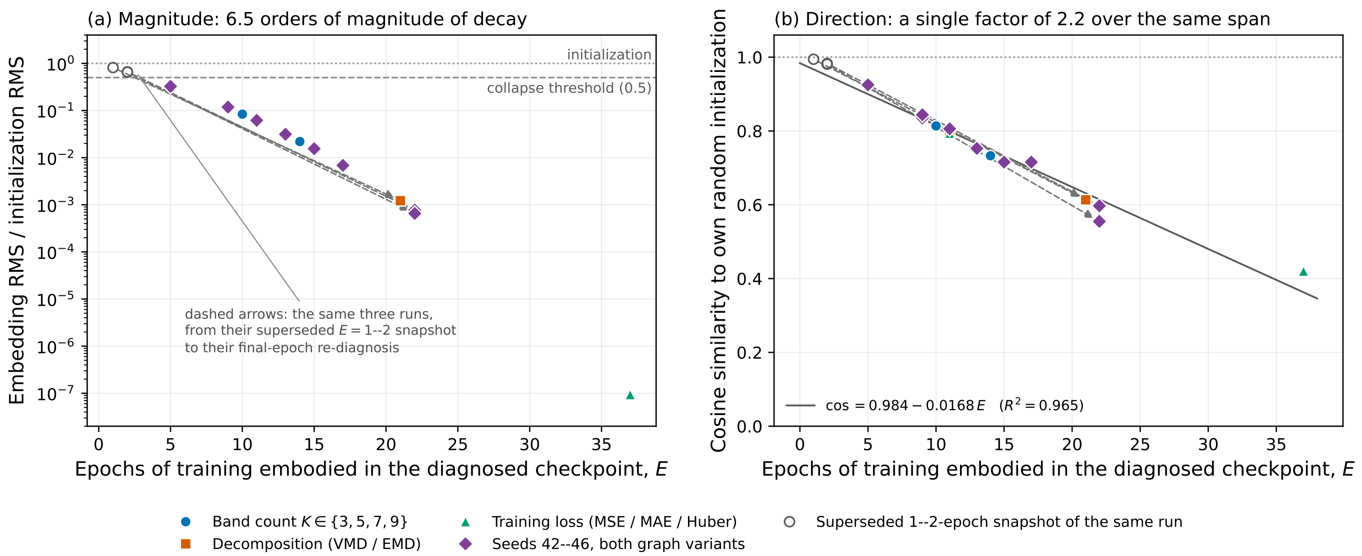

Fig. 5: The isotropic norm collapse, plotted from the 19 independently trained diagnostic cells of Section V-G (band-count sweep, decomposition comparison, loss comparison, and the five-seed repetition of both graph variants). The horizontal axis is the number of epochs of training embodied in the diagnosed checkpoint, E, which ranges from 5 to 37 across the suite. (a) Mean embedding RMS relative to the run's own random initialization, on a logarithmic scale: it falls from 0.331 at $E = 5$ to 9.4×10^{-8} at $E = 37$, crossing the diagnostic's 0.5 collapse threshold (defined against the Xavier reference RMS of 0.2885, which each run's own initialization matches to within 5.6% in every band plotted, so the two normalizations are interchangeable at this figure's scale) before the earliest cell in the corrected suite, and continuing monotonically thereafter (Spearman $\rho = -0.993$, $p = 3.4 \times 10^{-17}$). (b) Mean cosine similarity of the same embeddings to that same initialization, on a linear scale spanning the full $[0, 1]$ range: it moves only from 0.927 to 0.420 over the identical span, tracking training duration with $\cos = 0.984 - 0.0168\,E$ ($R^2 = 0.965$, $\rho = -0.990$). Marker shape and color identify which robustness axis produced each cell; the axes are visually interleaved, consistent with the absence of any residual axis effect once training duration is known (one-way ANOVA on the fit residuals, $F = 0.791$, $p = 0.518$). Open circles joined by dashed arrows are the three cells whose original best-validation checkpoints had been frozen by early stopping after one to two epochs and were subsequently re-diagnosed from final-epoch weights: these are the same three runs observed twice, and they traverse the same trajectory. Read together, the panels are the paper's central mechanistic claim — six and a half orders of magnitude of shrinkage against a factor of 2.2 of rotation, the signature of embeddings rescaled toward the origin by weight decay rather than rotated toward task structure by a learning signal. The residual rotation in (b) should not be read as learning: it is confounded with the shrinkage itself, since the cosine is computed on raw embedding vectors whose direction, at magnitudes of 10^{-3} and below, is increasingly dominated by accumulated floating-point residue rather than by any gradient — the gradient reaching these parameters is exactly zero in the three most-trained cells shown.

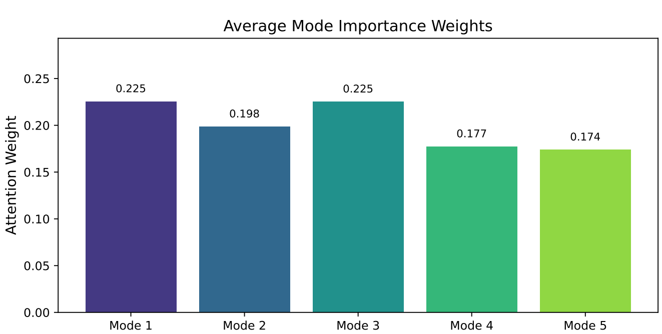

Fig. 6: Mode-fusion attention weights by VMD frequency band.

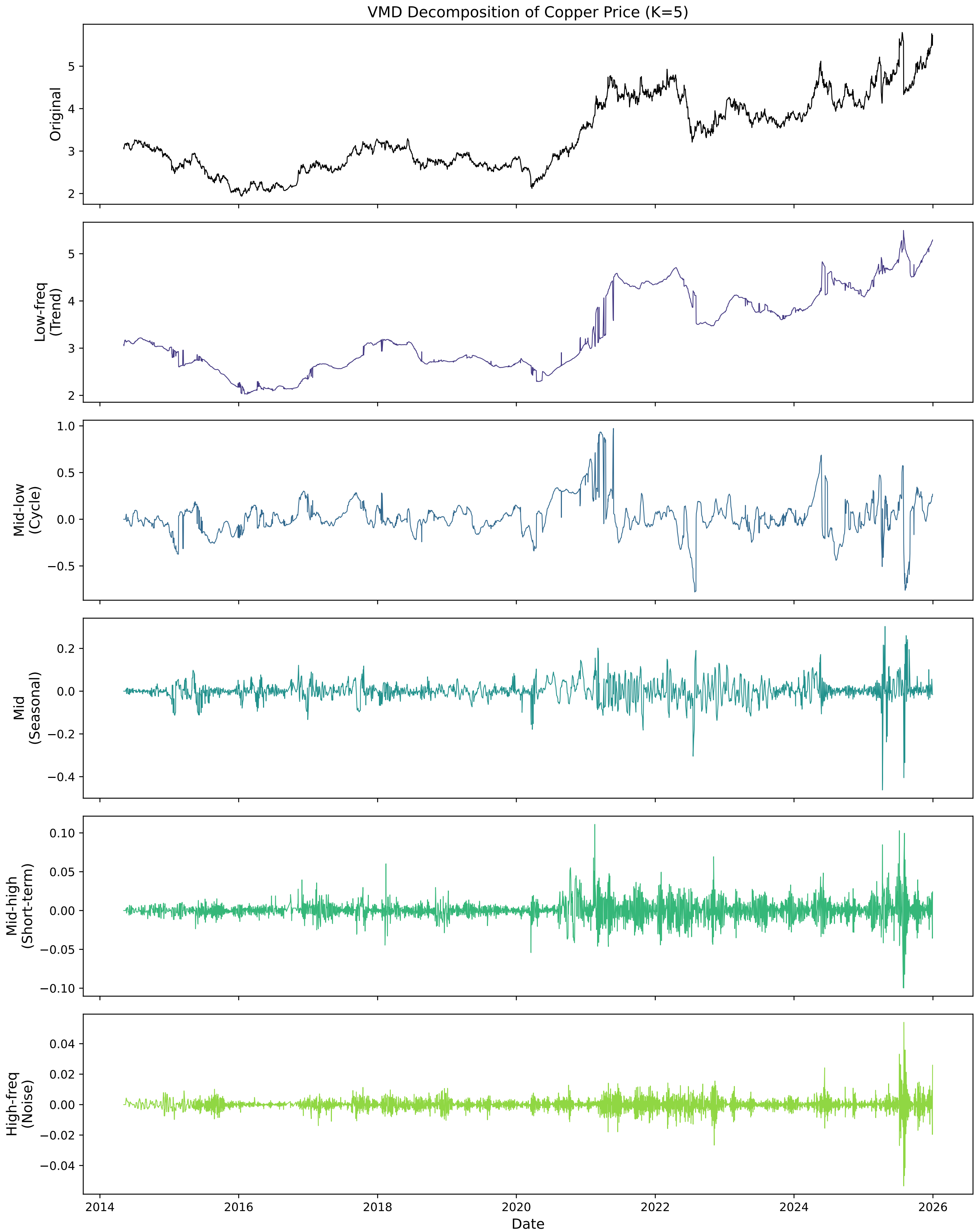

Fig. 7: Variational Mode Decomposition of the copper price series into $K = 5$ frequency-band modes.

F. Graph-Fix Experiment: Diagnosing, Fixing, and Re-Diagnosing the Collapse

Section V-E1's diagnosis identified two candidate mechanical causes of the adjacency collapse: coupled L_2 weight decay pulling every embedding parameter toward the origin, and the absence of any normalization on the bilinear logits before softmax, which lets embedding magnitude itself (rather than only direction) determine how peaked or flat the resulting adjacency is. A diagnosis that stops at "this is probably why" is weaker than one that tests the proposed mechanism directly, so we implemented and trained with two targeted fixes together: (1) L_2-normalizing $\mathbf{E}_k^{(1)}$ and $\mathbf{E}_k^{(2)}$ to the unit sphere immediately before the bilinear product, so the softmax input's scale can no longer collapse toward the origin regardless of what happens to the embeddings' raw magnitude; and (2) excluding the graph-embedding parameters from Adam's coupled weight decay entirely (a dedicated zero-decay parameter group), removing the constant, sign-consistent pull toward zero identified as the proximate driver of the collapse. Both fixes were applied together, not separately, since the diagnosis in Section V-E1 treats them as jointly responsible for the same failure mode. We retrained both `full_model_graphfix` (per-band) and `pooled_graph_graphfix` (pooled) with both fixes enabled, saved to `results_v2/`, keeping this run entirely separate from the original ablation (`results/`) and the tuned confound-check (Section V-D).

The fix mechanically works. Direct inspection of the trained `results_v2` checkpoints confirms the norm-collapse mechanism is resolved: embedding RMS magnitude is preserved at 0.96–1.01× its Xavier-initialization value across bands and models, in sharp contrast to the 1.4–3.7% of initialization observed in every unfixed checkpoint examined in this paper (Section V-E1, Section V-D). We independently re-verified this from the raw `results_v2` checkpoint state dictionaries rather than only citing the fix-experiment's own log: the pooled model's `graph_constructor.emb1/emb2` weight RMS (0.302 and 0.279) and the per-band model's five `graph_constructors.{0..4}.emb1/emb2` weight RMS values (0.27–0.30) both sit close to the 0.2885 Xavier-initialization value reported in Section V-E1, not the collapsed ∼0.009 value observed there. Whatever was driving the embeddings to zero in every earlier checkpoint no longer does so once weight decay is removed from these parameters and the bilinear product is computed on unit-normalized vectors.

The fixed mechanism still learns nothing. Mechanically preserving embedding norm is not the same as learning a functional graph, and it does not produce one here. The corrected embeddings moved only 0.6–2.0% in L2 norm away from their random initialization over the full training run, and retained cosine similarity of 0.9998–1.0000 with their own initialization—meaning the small amount of movement that did occur was almost entirely a rescaling within the same random-init direction, not a rotation toward any task-informed direction. The resulting adjacency matrices correlate with the adjacency computed from the *untrained*, randomly initialized embeddings at 0.994–0.999 per band. Put plainly: once norm collapse is no longer available as an explanation, the "fixed" per-band graph is statistically indistinguishable from a graph that was never trained at all. This confirms the deeper half of Section V-E1's diagnosis empirically, rather than only by inference: weight decay was not incidental to the collapse, it was the *only* force acting on these parameters in either direction, because the task gradient reaching them is too weak (on the order of 10^{-5} per element, as measured in Section V-E1) to move them meaningfully on its own. Removing weight decay's pull does not substitute a learning signal for it; it just leaves the embeddings approximately where they started.

One caveat applies specifically to `full_model_graphfix`'s own per-horizon metrics, if cited individually: this run early-stopped at epoch 20 under a patience of 20, meaning validation loss never improved past its very first epoch, so the saved checkpoint is effectively the epoch-0 initialization state, not a converged model, and its RMSE/MAE/DA numbers should not be read as characterizing a trained per-band model. `pooled_graph_graphfix` trained to a normal, substantially longer completed run and carries no such caveat; the embedding-collapse-resistance-without-learning finding above is drawn from both checkpoints' embedding geometry, not from either model's task-level accuracy, and is unaffected by `full_model_graphfix`'s early-stopping behavior.

We read this as strengthening, not weakening, the paper's account of the collapse: we diagnosed a specific mechanism, engineered a fix targeted precisely at that mechanism, confirmed the fix resolves the mechanism it was designed to resolve, and found that the architecture still fails to learn differentiated graph structure for a deeper, independently confirmed reason. A negative result that survives its own proposed remedy is more informative than one that was never tested against a remedy at all, and it sharpens the claim in Section V-E1 from "weight decay probably caused this" to "removing weight decay's pull does not fix it, because there was never enough task gradient here to learn from regardless."

G. Robustness Checks: Is the Collapse an Artifact of Our Configuration Choices?

Section V-E1's adjacency-collapse diagnosis and Section V-F's failed remedy were both established at a single configuration: $K = 5$ bands, VMD decomposition, MSE training loss, seed 42, and one set of tuned hyperparameters. A reader is entitled to ask whether the finding is a property of the architecture or an artifact of that particular corner of the configuration space. We therefore ran five robustness experiments — a band-count sweep ($K \in \{3, 5, 7, 9\}$), a decomposition-method comparison (VMD versus EMD), a training-loss comparison, a five-seed repetition of the two ablation variants carrying RQ1, and a substantially wider hyperparameter search — applying the *same* three-part gradient/adjacency diagnostic (embedding RMS relative to initialization, gradient norm reaching $\mathbf{E}_k^{(1)}/\mathbf{E}_k^{(2)}$, and cosine similarity against the run's own random initialization) to every one of the 19 trained cells. All 19 cells were trained at the tuned configuration of Section IV-C with weight decay 10^{-5}, and all 19 completed. Three of the cells were initially diagnosed from best-validation checkpoints that early stopping had frozen after one to two epochs of training; we identified this, added a code path that captures the final-epoch weights alongside the best-validation ones, and re-ran those three cells, so every diagnostic value reported below is measured on a checkpoint embodying at least five epochs of training. Two of the five experiments carry further defects we disclose in full below rather than repair, for reasons we state.

The collapse is invariant to every axis we varied, and is explained almost entirely by one variable we did not intend to vary. Across all 19 cells, each diagnosed on a checkpoint embodying between 5 and 37 epochs of training, the trained graph embeddings' cosine similarity to their own random initialization is predicted by that epoch count with Spearman $\rho = -0.990$ ($p = 7.5 \times 10^{-16}$); the embeddings' RMS magnitude relative to initialization gives $\rho = -0.993$ ($p = 3.4 \times 10^{-17}$), and the gradient norm reaching them gives $\rho = -0.957$ ($p = 1.4 \times 10^{-10}$). A linear fit in training epochs explains 96.5% of the variance in cosine-to-initialization ($\cos = 0.984 - 0.0168\,E$), a fit in $\log_{10}$ of the relative RMS explains 96.8%, and the residuals from the cosine fit show no detectable dependence on which experiment produced the cell (one-way ANOVA across the four axes: $F = 0.791$, $p = 0.518$; Kruskal–Wallis $p = 0.543$; largest absolute residual 0.060 against a full observed range of 0.420–0.927). Relative embedding magnitude falls from 33% of its initialization value in the least-trained cell (five epochs) to 9×10^{-8} of it in the most-trained (37 epochs; an absolute embedding RMS of 1.5×10^{-8} against a Xavier reference of 0.2885), passing through 1.2×10^{-3} at 21 epochs. Put plainly: once one knows how long a checkpoint trained, knowing whether it used three bands or nine, VMD or EMD, MSE or MAE, or which of five seeds, adds essentially nothing. This is the signature of a process driven by the optimizer rather than by the task, consistent with Section V-F's finding that removing weight decay's pull does not substitute a learning signal for it.

An early-stopping artifact we detected, corrected, and report because it is instructive. On a first pass, three of the 19 cells — the EMD arm, one seed of the per-band variant, and one seed of the pooled variant — were flagged `NOT COLLAPSED` by the automated diagnostic. They were not counterexamples. Our trainer saves the best-validation weights but stamps them with the loop's final epoch index; because the best-validation save condition and the early stopper's improvement condition are the same predicate (a strict improvement on the running best, with zero tolerance), and every run early-stopped well inside the 200-epoch budget, the weights actually in a checkpoint are exactly those from epoch (recorded epoch $-$ 20) under the patience of 20 used throughout. For these three cells the recorded indices were 0, 1 and 1, so their saved weights embodied one to two epochs of training against five to thirty-seven in every other cell, and their verdicts measured training duration rather than any property of the axis being varied. We therefore added a diagnostic path that retains the final-epoch weights alongside the best-validation ones and re-ran the three cells. All three reproduced their original early-stopping trajectory to the same final epoch and their original test metrics to within 2×10^{-6} relative, confirming that the re-diagnosis describes the same run.

All three collapse once diagnosed on genuinely trained weights. At 21, 22 and 22 epochs of training respectively, every band of all three cells is flagged collapsed on both of the diagnostic's independent criteria: the fraction of surviving adjacency edges within tolerance of $1/N$ is **1.00 in every band of all three**, and embedding RMS has fallen to 6×10^{-4} to 1.3×10^{-3} of the Xavier reference. The gradient norm reaching the embeddings is *exactly* zero for the EMD and per-band seed-44 cells and 2.7×10^{-22} for the pooled seed-45 cell, while the gradient reaching the rest of the network is 0.17–0.22 in the same measurement — a dead graph branch inside a live model. The three cells thus provide a within-run confirmation of the epoch relationship above that is stronger than any across-run comparison, because configuration, seed and training trajectory are all held fixed and only the snapshot moves: the same three runs read 0.994, 0.983 and 0.981 cosine-to-initialization at one to two epochs and 0.613, 0.597 and 0.555 at twenty-one to twenty-two. Two further cells re-run as a consistency check, which happened to train for 45 and 68 epochs, extend the same trajectory to relative embedding magnitudes of 1.4×10^{-10} and 3.2×10^{-22}.

We note for completeness that for these three cells the deployed (best-validation) checkpoint and the diagnosed (final-epoch) checkpoint are different weight sets. This does not soften the finding, because neither state contains learned relational structure: at the best-validation snapshot the embeddings are statistically indistinguishable from their random initialization (cosine 0.98–0.99, 65–82% of initialization norm), and at the final-epoch snapshot they are uniform with no gradient. The choice between the two is a choice between random and uniform.

With the correction in place, the collapse is observed in **19 of 19** runs, each diagnosed at a checkpoint embodying at least five epochs of training: at all four band counts, in both decomposition methods, in all three loss arms, and at all five seeds in each of the two graph variants. We state the epoch qualifier because it is load-bearing: the three re-diagnosed cells are the same runs whose one-to-two-epoch snapshots read `NOT COLLAPSED`, so what this suite establishes is not that the collapse is present at every epoch count, but that it is present in every run once as few as five epochs have elapsed, and that no axis we varied delays or prevents it. The decomposition-axis caveat we previously stated is withdrawn — EMD does not resist the collapse, and in fact, having trained ten epochs longer than the VMD cell, sits further along the identical trajectory (relative embedding magnitude 1.2×10^{-3} against VMD's 6.1×10^{-2}, and adjacency uniform to within 3×10^{-8} against VMD's 8×10^{-5}).

Accuracy differences across these axes are smaller than run-to-run noise. Four of the 19 cells are, by construction, the identical configuration ($K = 5$, VMD, MSE, seed 42) reached through four different experiment drivers, and we verified their held-out ground-truth arrays are bit-identical. Their test average MSE nonetheless spans 7.54% (0.002772–0.002981), which we take as this study's honest run-to-run noise floor at fixed configuration and fixed seed. Every axis effect we measured is smaller or comparable: the band-count sweep spans 8.1% (0.002757–0.002981), the loss comparison 9.4% (0.002691–0.002943), and the five-seed repetition of the per-band model has a coefficient of variation of 1.32% and a best-to-worst spread of 3.1%. We therefore report no accuracy ranking over K, decomposition method or loss; none is resolvable at $n = 1$ per cell. These four replicates also serve as a within-experiment check on the paragraph above: they embody 14, 11, 11 and 13 epochs of training and record cosine-to-initialization of 0.733, 0.794, 0.794 and 0.753, tracking training duration with configuration *and* seed both held fixed — which also shows that run-to-run nondeterminism perturbs the epoch count itself, and is why the configuration labels carry no residual information once training length is known.

The five-seed RQ1 check, and its capacity caveat. Repeating the two ablation variants that carry RQ1 across seeds 42–46 at the tuned configuration, the per-band model's test RMSE is statistically indistinguishable from the pooled variant's at every horizon (paired t-test over the five seeds: $p = 0.96$, 0.13, 0.88, 0.50 at $h = 1, 5, 10, 22$; Wilcoxon signed-rank likewise non-significant throughout). *This comparison is not capacity-matched.* It re-ran `pooled_graph_matched_dim`, which we confirm by direct parameter counting from the saved checkpoints carries 315,530 parameters against the per-band model's 1,461,124 — a 78.4% deficit — rather than the parameter-matched variant the RQ1 comparison calls for. That 78.4% figure is *not* the same measurement as the 78.5% reported in Section III-G: the latter is a different model pair, at the untuned default configuration (hidden dimension 64), and the two land 0.1 percentage points apart by coincidence. We report the comparison anyway, in the conservative direction it happens to run: a per-band model with $4.6\times$ the parameters does not significantly beat a much smaller pooled model at any horizon across five seeds. This strengthens rather than weakens Section V-I's negative answer to RQ1, and we note that the parameter-matched version of this comparison is one this paper retracts on independent grounds in Section V-H, so there is no live claim here that the capacity mismatch could undermine. Two of the ten multi-seed cells (per-band seed 44, pooled seed 45) are among the three re-diagnosed from final-epoch weights described above. Both re-ran to the same final epoch as originally recorded and reproduced their test metrics to within 10^{-8} absolute, so the paired tests reported here are unchanged; only their collapse verdicts change, from `NOT COLLAPSED` to `COLLAPSED`, which is what brings the per-seed coverage to all five seeds in both variants.

The loss comparison tested two losses, not three. We configured MSE, MAE and Huber arms. The Huber arm used PyTorch's `smooth_l1_loss` at its default $\beta = 1.0$, which is quadratic — exactly $\frac{1}{2}x^2$ — for residuals below 1.0. The largest absolute residual observed anywhere in this experiment, across all three arms and all four horizons, is 0.284, i.e. 28.4% of β; the loss therefore never once entered its linear branch, and the Huber arm is mathematically identical to $\frac{1}{2}\times$MSE. We report it as a two-loss comparison and do not claim a three-way result. The arm is not uninformative, however: because Adam's coupled L_2 decay does not scale with the loss, halving the task gradient while leaving the decay pull intact doubles the decay's relative influence, and this arm consequently trained longest (37 epochs of retained weights, against 11 for the MSE arm) and collapsed hardest, terminating with embedding RMS of 1.5×10^{-8} against a Xavier reference of 0.2885, a bit-exactly uniform adjacency (softmax standard deviation of exactly zero at band 0), and a gradient reaching the embeddings of *exactly* zero. An unintended loss-scale ablation thus independently corroborates the weight-decay mechanism argued for in Sections V-E1 and V-F.

A wider hyperparameter search does not rescue the model, and confirms the original search was adequate. The study reported in Section IV-C searched five parameters over 10 trials at fixed $K = 5$, batch size 32 and weight decay 10^{-5}. We re-ran a study three times larger and with three additional dimensions — $K \in \{3, 5, 7, 9\}$, weight decay (log-uniform, 10^{-6} to 10^{-3}) and batch size $\in \{16, 32, 64\}$ — for 30 trials under an identical objective (best validation average MSE over 25 epochs). Twelve trials completed and 18 were pruned. The best configuration found (hidden dimension 128, 8 heads, learning rate ≈ 0.00277, dropout ≈ 0.0904, 2 GNN layers, weight decay $\approx 2.0 \times 10^{-4}$, batch size 16, $K = 9$) attains validation MSE 0.0020145 against the deployed configuration's 0.0020509 — an improvement of **1.78%**, with only 2 of the 12 completed trials beating the deployed value at all, and the second-best tying it to four significant figures. That margin is smaller than this study's 3.1% across-seed spread and far smaller than its 7.54% same-configuration replicate spread, and part of it is attributable to optimizer step count rather than configuration quality, since the winning trial halves the batch size within a fixed epoch budget. We therefore report the wider search as a confirmatory negative: the original five-parameter, ten-trial configuration was already effectively optimal for this objective, and we do not re-run the main results at the new one. Three caveats attach. First, all 18 pruned trials were pruned after a single epoch, the median pruner having been constructed without warmup steps, so 60% of the search budget was screened on one epoch of validation loss, which structurally favors the high-learning-rate, small-batch region the survivors occupy. Second, the sampler concentrated on $K = 9$ (8 of the 12 completed trials), and batch size is fully aliased with K in the completed set, so the per-parameter marginals and importances are confounded and we do not interpret them; independently, the band-count sweep's converged test results do not reproduce the search's implied K ordering, and span less than the noise floor. Third, and most consequential for this paper's argument, **the search objective is structurally blind to the collapse**: it scores validation MSE and nothing else, and in our 19 diagnosed cells the near-initialization checkpoints score squarely mid-range among the collapsed ones (training length correlates with test average MSE at only $\rho = 0.456$, $p = 0.05$). Collapse is essentially free under this objective, so no widening of the search space could select against it. We did not capture embedding diagnostics for individual search trials and therefore cannot state directly whether the collapse persists at the wider optimum; we note only, as inference from the established mechanism rather than as measurement, that the selected configuration's weight decay is 20 times the value used everywhere else in this paper, which points toward more severe collapse rather than less.

Taken together, these checks close the most obvious escape route from Sections V-E1 and V-F's negative findings. The collapse is not a property of $K = 5$, of the squared-error loss, of a single seed, or of an insufficiently searched hyperparameter space; whether it is a property of VMD specifically remains open pending the re-diagnosis flagged above, since the EMD arm is the one cell this suite cannot currently speak to. What the 16 usable cells show is that the collapse is a property of what happens to these embeddings under the optimizer once training proceeds past its first epoch, and that it does not depend on any modeling choice we were able to vary.

H. The Constant-Forecast Artifact: Why Directional Accuracy Needs a Dispersion Check

While investigating the `results_v2` graph-fix checkpoints' predictions directly, in order to understand `pooled_graph_graphfix`'s directional-accuracy numbers, we found something that changes how several directional-accuracy comparisons elsewhere in this paper should be read. `pooled_graph_graphfix`'s predicted returns have standard deviation ranging from 1.11×10^{-5} ($h = 1$) to 6.33×10^{-4} ($h = 22$) across the test set, against a true-return standard deviation of 1.79×10^{-2} to 7.58×10^{-2} over the same horizons—three to four orders of magnitude less dispersion than the target. Directly inspecting the raw prediction arrays shows why: at every one of the four horizons, this model's 984 test-set predictions are of a single, constant sign (all negative). A model whose forecast never changes sign is not exercising directional judgment on any individual sample; its "directional accuracy" can only ever equal the fraction of test days on which the true return happened to share that one sign.

We verified this precisely rather than assuming it: recomputing directional accuracy directly from the saved `results_v2/predictions/pooled_graph_graphfix_h{1,5,10,22}.npy` prediction and true-value arrays reproduces the reported DA figures (48.374, 46.951, 44.207, 44.715 at $h = 1, 5, 10, 22$) to four decimal places, and each of these values in turn equals, to four decimal places, the test period's unconditional frequency of the sign this model always predicts (down-day frequency 48.374%, 46.951%, 44.207%/44.614% depending on zero-return handling, and 44.715% at the respective horizons). This is not a coincidental resemblance; it is an identity, because a constant-sign forecaster's accuracy is mathematically the base rate of the sign it happens to emit, no more and no less. The same pattern reproduces in `pooled_graph_matched_params_tuned` from the confound-check experiment (Section V-D): its four DA values (48.37, 53.05, 55.39, 44.72) match the corresponding test-period class frequencies to the same precision, even though we did not archive that run's raw prediction arrays and can only confirm it via this frequency match rather than by inspecting prediction dispersion directly. Cross-referencing further, this pattern is consistent with what motivated the `results_v2` investigation in the first place: multiple models across the original untuned ablation (Section V-C), the tuned confound-check (Section V-D), and the fix experiment above show DA values that coincide with plausible class-frequency figures for their horizon, though `pooled_graph_graphfix` and `pooled_graph_matched_params_tuned` are the two cases we can state this about with full confidence, having checked prediction dispersion or exact frequency-matching directly. A binomial test of each model's hit count against its horizon's majority-class frequency, run across every model$\times$horizon cell in the fix-experiment run, fails to reject the majority-class null at conventional significance levels in every cell tested—these apparent directional-accuracy figures do not represent a statistically demonstrated departure from what the majority class alone would produce.

The direct consequence is a retraction. Section V-D reported that, at HPO-tuned hyperparameters, `full_model_tuned` beats `pooled_graph_matched_params_tuned` on directional accuracy at three of four horizons, which on its face reverses this paper's negative ablation finding and would suggest a real per-band advantage once capacity and tuning are no longer confounded. **We do not carry that reading forward, and explicitly retract it here.** The comparison's losing side, `pooled_graph_matched_params_tuned`, is the same class of degenerate constant-sign forecaster documented above; its DA values are an artifact of which constant sign it happened to emit relative to each horizon's majority class, not a measurement of directional skill it lost to a better-performing per-band model. A win recorded against a degenerate opponent is not evidence of skill on the winning side either. We therefore withdraw any claim, hedged or otherwise, that per-band graph construction demonstrates a real directional-accuracy advantage over pooling at this data scale; no experiment in this paper supports that claim once prediction dispersion is checked, and Section V-I's answer to RQ1 is written accordingly.

We report this as a general methodological finding, not only a correction specific to this paper's own results, because nothing about a bare directional-accuracy percentage would have revealed it: 48.37% or 44.72% both look like unremarkable, plausible directional-accuracy figures for a financial return-forecasting model in isolation, and would have passed an ordinary reviewer's sanity check without triggering suspicion. We arrived at this check independently, while auditing our own tables, but we claim no priority for the underlying concern: Cheung (2026), in work posted shortly before we identified the artifact in our own results, makes the same core argument for equity forecasting with a LoRA-adapted time-series foundation model, showing that an apparent 80% directional accuracy was the base rate of a trivial "always-up" rule in a rising market, and proposing excess accuracy over that base rate — with paired McNemar and Diebold-Mariano tests — as the headline directional metric. Recommendation (b) below is that same recommendation, and we cite it as prior work rather than restate it as ours. What this paper adds is on the degeneracy-detection side rather than the reference-point side: (i) the sign-concentration statistic, which establishes that the *model* is a constant-sign forecaster and not merely that the null was misspecified; (ii) the base-rate-*identity* check, which requires only a reported DA figure and the evaluation period's class frequency, and can therefore be applied retroactively across a grid of already-reported results whose raw prediction arrays were never archived, as we do for every cell of Table III; and (iii) the explicit counterexample below showing that a prediction-dispersion check does not substitute for either. The artifact was only caught by explicitly checking prediction sign concentration and comparing directional accuracy against the test period's own majority-class frequency rather than a generic 50% null—two checks that are cheap to run but not part of how directional accuracy is conventionally reported in this literature, including in this paper's own earlier drafts. We recommend, as a general practice for directional-accuracy reporting in financial forecasting, that it always be accompanied by (a) the *fraction of predictions sharing the majority sign*, combined with a binomial test of the hit count against the majority-class null, so a near-constant-sign forecaster is visible on inspection, and (b) a comparison against the evaluation period's own majority-class frequency rather than an assumed 50% baseline, since a period with meaningful directional drift (as this study's 2022–2025 test period has, Section V-A) makes 50% the wrong null to test against regardless of whether any individual model is degenerate. We deliberately do not recommend prediction standard deviation as this check, even though it is a natural first instinct: this paper's own main results table (Table I) contains a direct counterexample. `Transformer`'s $h = 5$ predictions have a prediction-to-true standard deviation ratio of 0.42—healthy, unremarkable dispersion by the standard a variance-only check would apply—yet 100% of those predictions share the same sign, because a mean shift larger than that spread pushes every prediction positive regardless of how much it varies around that shifted mean. A dispersion check alone would not have flagged `Transformer_h5` as degenerate; only the sign-concentration check does. A directional-accuracy number reported without the sign-concentration and majority-class-null checks is not, by itself, evidence of directional skill.

I. Discussion: Answering the Research Questions

RQ0 (does any of this beat doing nothing?), asked first because it conditions how RQ1–RQ3 should be read. No. Table I shows the naive zero-forecast baseline wins MAE at all four horizons against all seven fitted models, and RMSE at two of four; every model's R^2 is negative or near zero. This is the most important finding of this study and the reason RQ1–RQ3 below should be read as differences *among* approaches that do not clearly explain copper return variance, not as evidence that any of them, including VMD-MFGNN, has found exploitable signal.

RQ1 (per-band versus pooled graph). Not supported, now explained mechanistically (Section V-E1, Section V-F) and checked for a directional-accuracy artifact that could otherwise have overturned this answer (Section V-H). Section V-C's precise reading of Table III is that the per-band full model never attains the best-of-six value in any column, and loses directional accuracy to both pooled-graph variants at three of the four horizons ($h = 1, 5, 10$) — its one DA win over a pooled variant coming at $h = 22$ against `pooled_graph_matched_dim` only, a variant carrying 78.5% fewer parameters — while trading RMSE/MAE wins back and forth with them at individual horizons. Given the adjacency-collapse diagnosis in Section V-E1 — every per-band adjacency matrix converged to $\approx 1/N$ uniform weights, with two independently trained models sharing the same collapsed, undifferentiated embedding direction — this negative ablation result is not a surprising or mysterious finding about graph topology; it is close to the expected outcome once the per-band mechanism carries almost no differentiated information to begin with. We therefore answer RQ1 in the negative for a stronger, more specific reason than "the ablation numbers came out this way": the per-band graph mechanism, as trained here, did not learn the kind of structure that could produce a per-band advantage, so its absence in the ablation table is consistent with, not contradictory to, the mechanistic finding. A tuned re-check (Section V-D) initially appeared to overturn this answer, with the per-band model beating the pooled variant on directional accuracy at three of four horizons; we traced that apparent reversal to the pooled variant emitting a near-constant, near-zero-variance forecast whose directional accuracy exactly reproduces the test period's class frequency rather than measuring real skill (Section V-H), and we explicitly retract the reversal rather than let it stand as a competing answer to RQ1. We engineered and tested a targeted fix for the adjacency collapse itself (Section V-F): it mechanically resolves the collapse (embedding norm is preserved) but the embeddings still fail to move meaningfully from random initialization under the task gradient available to them, so RQ1's negative answer holds under every configuration and every diagnostic check we applied, not only the original untuned ablation. Section V-G extends that statement across four further axes: the collapse recurs at every band count, in every loss arm, and at every seed of both graph variants that trained beyond a first epoch, and a three-times-wider hyperparameter search neither improves the model meaningfully nor could select against the collapse, since its objective is validation error alone. A five-seed repetition of the two RQ1 variants at tuned hyperparameters likewise finds no significant per-band advantage at any horizon, in a comparison whose capacity mismatch runs in the conservative direction.

RQ2 (per-band graph versus no-VMD and versus a fixed correlation graph). Also not cleanly supported. Against the no-VMD, matched-params variant, the full model loses on RMSE and MAE at $h = 1$ (the no-VMD variant achieves the best RMSE/MAE of all six ablation variants at that horizon) while winning at $h = 22$ on RMSE only (the DA-at-$h = 22$ "worst of six" claim for that variant does not hold, Section V-C); the picture is mixed rather than a decomposition advantage at every horizon. Against the correlation-graph variant, which isolates learned versus fixed adjacency while holding the per-band architecture fixed, the correlation graph wins the best RMSE of all six variants at $h = 22$ and is close to or better than the full model at most other horizon/metric combinations — itself a natural consequence of the collapse finding: a fixed correlation graph is, almost by construction, no worse a source of relational structure than a learned graph that failed to learn any. We answer RQ2 in the negative as well: neither the contribution of VMD decomposition nor the contribution of end-to-end learned (versus fixed correlation-based) adjacency is cleanly demonstrated by these ablations.

RQ3 (statistically significant edge over standard baselines). Answered with a qualified yes, more qualified after multiple-testing correction. At uncorrected $\alpha = 0.05$, the HPO-tuned VMD-MFGNN (Table II) achieves a significant Diebold-Mariano advantage over four of six baselines — LSTM (2 of 4 horizons), Transformer (3 of 4), SimpleMTGNN (3 of 4), VMD-LSTM (4 of 4) — 12 of 24 comparisons overall, plus one significant loss (XGBoost at $h = 1$). After Holm-Bonferroni correction across all 24 tests, 8 of 24 comparisons remain significant, all favoring VMD-MFGNN; the XGBoost loss does not survive correction. VMD-MFGNN remains statistically indistinguishable from ARIMA at every horizon, corrected or not. The qualified answer to RQ3 is: yes, a real, correction-surviving edge exists against the deep-learning baselines architecturally closer to VMD-MFGNN's design space, but no significant edge, in either direction, is demonstrated against the two simpler, non-deep-learning baselines, this comparison is further qualified by the disclosed HPO asymmetry (VMD-MFGNN was tuned, none of the six baselines were), and, per RQ0, a significant DM result establishes only that two models' errors differ, not that either has found real structure in copper returns.

Taken together, the honest overall picture is: the HPO-tuned VMD-MFGNN is a competitive forecaster relative to other deep-learning approaches, is not clearly worse than ARIMA or XGBoost, and its significant advantages survive a stricter multiple-testing correction, but (a) none of this changes the fact that it, and every other model tested, fails to beat a trivial zero-forecast baseline on MAE (RQ0), and (b) this study's ablations do not provide architectural evidence that the specific mechanism motivating the paper — per-band graph construction over a single pooled graph — is what drives whatever competitiveness against other fitted models does exist, for a reason we can now name precisely and have since tested directly: that mechanism did not learn differentiated graph structure in the trained model this paper reports, a targeted fix for the diagnosed collapse mechanism does not change this (Section V-F), and an apparent directional-accuracy reversal that briefly suggested otherwise turned out to be a constant-forecast artifact rather than a real result (Section V-H). We view the combination of RQ0's naive-baseline result, the mechanistic diagnosis and verified fix attempt behind RQ1, and the constant-forecast methodological finding as the central, disclosed empirical findings of this study, not footnotes to a more triumphant architectural narrative.

J. Economic Significance

A model can show a statistically significant DM advantage over another forecaster while still carrying no economic value, and RQ0's naive-baseline result already suggests that is the more likely situation here; we check this directly with a lightweight, unlevered long/short economic sanity check (position sized by predicted sign, non-overlapping-period annualization, no explicit transaction-cost model beyond the reported breakeven threshold) computed post hoc from the saved test-set predictions. Across all seven fitted models and four horizons, annualized Sharpe ratios computed this way are inconsistent in sign across horizons for every model, including VMD-MFGNN (ranging from about -0.45 to $+1.14$ across the 28 model/horizon combinations, with every model switching sign at least once across horizons); breakeven round-trip transaction costs implied by these Sharpe ratios are, for over half of the model/horizon combinations, in the single-digit basis points or negative, well inside or below realistic copper-futures round-trip transaction costs, and Pesaran-Timmermann directional-value tests are not significant at conventional levels for the large majority (21 of 28) of model/horizon combinations. We read this as corroborating RQ0 rather than adding a new finding: consistent with no model beating a trivial zero-forecast baseline on MAE, no model in this study demonstrates directional forecasting value robust enough to survive realistic trading frictions, and the occasional positive Sharpe ratio at a single horizon for a single model should be read as noise around zero rather than as evidence of exploitable skill, given how frequently sign and magnitude both flip across adjacent horizons for the same model. VMD-MFGNN's own $h = 1$ cell is the largest single value in this range (annualized Sharpe 1.14, breakeven cost 233.5 bps) and is worth flagging rather than passing over quietly: even so, its underlying Pesaran-Timmermann directional-value test is not significant, raw or Holm-corrected ($p = 0.090$), and its DA edge over the majority-class null is only 1.5 percentage points (Section V-A), so we read this single cell as within the noise this section otherwise describes rather than as a standout economic result.

VI. Limitations

We enumerate the limitations of this study's design explicitly, rather than deferring them to a brief closing paragraph.

1) **Zinc and nickel dropped from scope entirely.** Zinc and nickel were originally planned inputs, sourced via NASDAQ commodity sub-index proxies (^NQCIZNER, ^NQCINIER) since neither has a standalone Yahoo Finance futures contract (both are LME-only instruments). On the live production data pull for this study, both proxy tickers were confirmed dead/delisted, with no further viable Yahoo Finance substitute identified, so both variables were removed from scope entirely rather than replaced. This is a deliberate, disclosed reduction from an originally planned 10-variable system to the 8-variable system used throughout this paper ($N = 8$: copper, aluminum, gold, oil, DXY, S&P 500, VIX, US 10-year yield); results and interpretability figures reflect this narrower set of inter-commodity relationships, and any relationship specifically involving zinc or nickel cannot be studied with this dataset.
2) **VMD rolling-window length and daily refit cost.** The rolling-window VMD decomposition (Section III) is refit

at every trading day on the trailing 252-day window, which removes the staircase/lag issue an earlier, since-abandoned 21-day-refit design would have had, but at a real computational cost: full daily-refit VMD across all $N = 8$ variables and the full 2010–2025 date range takes on the order of 30 minutes of single-core CPU time (one-time, then cached to disk). The 252-day window length itself is a fixed choice, not tuned; a shorter or longer window could change the resulting mode boundaries and was not swept within this project's timeline.

3) **No purge or embargo at split boundaries.** The train/validation/test split boundaries (Section IV) are simple calendar cutoffs; samples whose h-day-ahead target crosses a split boundary (up to $h = 22$ trading days) are not purged or embargoed from the adjacent split, a standard precaution in some financial ML protocols that this study does not implement. Given the small size of any single leakage window relative to each split's multi-year length, we judge the practical impact to be small, but do not claim it is zero.
4) **Non-stationary price-level inputs and covariate shift.** Several baselines and the no-VMD ablation variant (Section IV-D) consume price-level (not VMD-decomposed) inputs, standardized using statistics computed on the 2010–2019 training period. Copper and several of the other seven input series moved well outside their 2010–2019 range by the 2022–2025 test period, so these models face meaningful covariate shift at test time that a return-based or rolling-normalized input representation would reduce; this was not corrected within this project's timeline.
5) **Continuous-futures roll-gap contamination.** Several input series (notably crude oil, which recorded a well-known negative-price episode in April 2020, inside this study's validation period) are continuous futures series that can contain roll-related price discontinuities not present in the underlying economics. No explicit roll-gap screening or outlier capping was applied to any input series.
6) **Single chronological split; no walk-forward evaluation.** All results are computed on one train (2010–2019) / validation (2020–2021) / test (2022–2025) split, not a walk-forward or rolling-origin design with multiple folds. This limits the results to a single realization of test-period market conditions, and, combined with the absence of multiple seeds in that run, means there is no resampling unit available to compute a paired significance test or effect size (e.g., Cohen's d) for the ablation comparisons (Table III); only the baseline comparison (Table II) admits a valid Diebold-Mariano test, because it compares two forecast-error series over the same single test period rather than requiring resampling across folds or seeds. The five-seed repetition reported in Section V-G supplies a seed-level resampling unit for the two RQ1 variants specifically, at the tuned configuration, but it is still a single chronological split; it does not address the fold-level limitation described here, and no walk-forward evaluation is performed anywhere in this paper.
7) **Asymmetric hyperparameter optimization, and fewer trials than configured.** VMD-MFGNN's HPO search (Section IV-C) was configured for up to 15 Optuna trials but, in the run reported here, completed only 10 (5 `COMPLETE`, 5 `PRUNED`) before the search budget was exhausted; the reported best configuration is the best of those 10, not necessarily the best of a full 15-trial search. The six baselines receive no HPO at all and run at standard, hand-set configurations. Both facts are deliberate, disclosed choices/outcomes made to fit the project's compute and time budget, not oversights, but they mean the baseline comparison (Table I, Table II) compares a tuned model, tuned over fewer trials than originally configured, against six untuned ones, and any accuracy gap is not solely attributable to architecture. On the first of these two points specifically, a later, substantially wider search — 30 trials over eight parameters, adding K, weight decay and batch size — found no meaningful improvement over the reported configuration (1.78% in validation MSE, inside this study's own replicate noise floor), which we report in Section V-G as evidence that the truncated original search was not the binding constraint. It does not address the second point: the six baselines remain untuned.
8) **Ablation results do not transfer directly to the HPO-tuned model, and do not support the paper's central claim at the configuration tested.** All six ablation variants (Section IV-D), including the ablation study's own `full_model` row, are trained at the untuned default configuration (393,476 parameters), a different and much smaller trained instance than the HPO-tuned VMD-MFGNN reported in the main comparison (1,461,124 parameters, Section III-G). Capacity matching for the no-VMD and pooled-graph variants worked well in absolute terms (matched-parameter variants land within about 1% of the full model's parameter count), which resolves the raw capacity-confound concern that motivated building two capacity settings per variant; both the matched-dimension and matched-parameter settings are reported at genuinely matched capacity, and no ablation row should be read as carrying a residual, unaddressed capacity confound relative to the others. Resolving the capacity confound did not, however, resolve the substantive architectural question it was meant to isolate: even with capacity approximately matched, the per-band full model never attains the best-of-six value in any RMSE/MAE/DA column in Table III, and the capacity-matched pooled-graph variant beats it on directional accuracy at three of the four horizons ($h = 1, 5, 10$), the fourth ($h = 22$) being a confirmed base-rate artifact we do not count in either direction (Section V-C, Section V-H). This is reported as a substantive limitation on the paper's central architectural premise, not merely a methodological caveat to be read past, and Section V-I argues it is the expected consequence, not a separate puzzle, given the adjacency-collapse finding below.
9) **The learned per-band adjacency collapsed to near-**

uniform edge weights; the interpretability evidence supports no claim of learned relational structure, and this persists after a targeted fix. Direct inspection of trained checkpoints (Section V-E1) shows every kept edge weight in every one of the five VMD frequency bands equal to $\approx 1/N = 0.125$ to five decimal places, traced to isotropic norm collapse of the underlying graph embeddings (RMS magnitude down roughly 32-fold from initialization) under the interaction of coupled weight decay and a weak task gradient. Two independently trained models' embeddings differ only by scale (cosine similarity > 0.99), indicating no directional/relational learning occurred at all, not merely that the learned structure is subtle or hard to visualize. We subsequently engineered and tested a fix targeting this exact mechanism — unit-normalizing the embeddings before the bilinear product and excluding them from weight decay (Section V-F) — and confirmed it resolves the norm collapse (embedding RMS preserved at 0.96–1.01× initialization) without resolving the underlying failure to learn: the corrected embeddings move only 0.6–2.0% in L2 norm from random initialization, retain cosine similarity 0.9998–1.0000 with their own initialization, and produce adjacency matrices correlated 0.994–0.999 with the untrained, randomly initialized adjacency. This is now the primary interpretability finding of this paper and should be read together with the ablation limitation above and with the constant-forecast finding (Section V-H): the per-band mechanism's failure to demonstrate an accuracy advantage (Section V-C), its failure to learn differentiated structure even before any fix (Section V-E1), and its continued failure to learn differentiated structure after a verified, targeted fix (Section V-F) are the same underlying phenomenon viewed from three angles, not three independent negative results, and an apparent tuned-configuration reversal of the ablation finding (Section V-D) does not survive scrutiny either, once traced to a degenerate constant-forecast comparison (Section V-H). The mode-fusion attention weights are separately near-uniform (0.19–0.22 across bands), but that artifact contains only 24 test-set samples and does not support a claim about drift or trend over the test period.

10) **Directional accuracy is not interpretable without a sign-concentration and majority-class-null check, and this paper's own earlier drafts did not apply one.** Section V-H documents that at least two models evaluated in this study (`pooled_graph_graphfix` and `pooled_graph_matched_params_tuned`) emit near-constant, near-zero-variance forecasts whose directional accuracy exactly reproduces the test period's unconditional class frequency at each horizon rather than reflecting real directional skill, and that this artifact was responsible for an apparent reversal of this paper's central ablation finding that we have since retracted (Section V-D). We did not archive per-sample prediction arrays for every ablation and confound-check variant, so we cannot directly check prediction dispersion or sign concentration for every reported DA figure; where those raw arrays are unavailable, we instead applied the base-rate-identity half of the check (DA value against the test period's class frequency, Section V-H) to every cell of the untuned ablation table (Table III) directly, since that check requires only the DA figure and the known class frequencies and not the underlying predictions. This found six confirmed artifact cells, marked in Table III (Section V-C); RMSE/MAE comparisons, which are not subject to the same constant-forecast artifact, should still be weighted more heavily than DA comparisons wherever the two are in tension.
11) **Multiple testing across 24 significance tests.** The baseline comparison runs 24 Diebold-Mariano tests (six baselines × four horizons); we apply a Holm-Bonferroni correction at family-wise $\alpha = 0.05$ (Section IV), which reduces the significant count from 12 to 8 (all favoring VMD-MFGNN) and removes the one significant result favoring a baseline (XGBoost at $h = 1$). We report both raw and corrected significance in Table II rather than only the more favorable raw count.
12) **Single-seed main and ablation results; seed variance characterized only in a separate robustness run.** Every number in Table I, Table II, Table III and Table IV comes from a single training run per model/variant and should be read as a point estimate rather than as characterizing a distribution; we did not repeat those runs across seeds. Seed-to-seed variance *is* characterized, but only for the two variants carrying RQ1 and only at the tuned configuration, in the five-seed robustness experiment of Section V-G, where the per-band model's test average MSE has a coefficient of variation of 1.32% across seeds. That experiment additionally established a larger and more troubling figure: four cells at a genuinely identical configuration and identical seed, differing only in which experiment driver launched them, span 7.54% in test average MSE. Run-to-run nondeterminism at this data scale therefore exceeds every architectural axis effect this study measured, which is the correct scale against which to read any single-run difference reported in the tables above.
13) **Three robustness cells were initially diagnosed from barely-trained checkpoints; the diagnosis has been corrected and we report the correction rather than suppress it.** Three of Section V-G's 19 diagnostic cells (the EMD arm, the per-band seed-44 run, and the pooled seed-45 run) early-stopped such that their saved best-validation weights embodied only one to two epochs of training, so their automated "not collapsed" verdicts measured training duration rather than any property of the axis being varied. We added a diagnostic path that retains each run's final-epoch weights alongside its best-validation weights and re-ran those three cells; all three reproduced their original trajectories to the same final epoch and their original test metrics to within 2×10^{-6} relative, and all three are collapsed when diagnosed at 21–22 epochs of training, with every band's surviving adjacency edges uniform to within tolerance of $1/N$ and

the gradient reaching the graph embeddings exactly zero or of order 10^{-22}. The collapse claim in that section is therefore stated over 19 of 19 runs, each diagnosed at a checkpoint embodying at least five epochs of training, rather than over 16 of 16 with three cells set aside; the qualifier matters, since the three re-diagnosed runs are the same ones whose one-to-two-epoch snapshots read "not collapsed", so what the suite establishes is that the collapse appears in every run within a few epochs and that no axis we varied delays it, not that it is present at every epoch count. The decomposition axis now carries usable evidence in both directions. Two residual caveats remain. First, for these three cells the deployed best-validation checkpoint and the diagnosed final-epoch checkpoint are different weight sets; we did not measure the final-epoch weights' test accuracy, and we rest the claim on the observation that neither snapshot contains learned relational structure — the earlier one is indistinguishable from random initialization (cosine 0.98–0.99 to its own init) and the later one is uniform with no gradient. Second, the decomposition axis still rests on a single cell per method, so it establishes that EMD does not *avoid* the collapse but does not support any accuracy comparison between the two decompositions.

14) **Asymmetric hyperparameter optimization interacts with, but does not explain, the naive-baseline result.** The HPO asymmetry disclosed above (item 7) means the main-table comparison among the seven fitted models is not a fully controlled architecture comparison. It does not, however, offer an explanation for why every model, tuned or not, fails to beat the naive zero-forecast baseline on MAE (Section V-A): additional tuning of the baselines would be expected to narrow gaps *among* the seven fitted models, not to change whether any of them exceeds a baseline that requires no tuning at all.

VII. Conclusion

Section V-I answers this paper's research questions in full against the real results; we do not re-derive those answers here, only summarize them. The naive-baseline result (RQ0) is the finding that should anchor how the rest is read: none of the seven fitted models, including VMD-MFGNN, beats a trivial zero-forecast baseline on MAE at any horizon. Against that backdrop, RQ1 and RQ2 (per-band versus pooled graphs, and versus no-VMD/fixed-correlation alternatives) are answered in the negative, and, unlike a typical negative ablation result, we can say why, and we have since tested that explanation directly rather than resting on it: the per-band graph-learning mechanism collapsed to near-uniform edge weights during training (isotropic embedding norm decay under coupled weight decay, Section V-E1); a targeted fix removing that exact decay pull and normalizing the embeddings mechanically resolves the collapse but leaves the embeddings essentially at their random initialization, confirming the deeper cause is a task gradient too weak to learn from, not merely an unlucky optimizer interaction (Section V-F); and a tuned re-check that briefly appeared to overturn RQ1 traced to a degenerate, near-constant forecaster whose directional accuracy was a class-frequency artifact rather than real skill, which we explicitly retract (Section V-D, Section V-H). RQ3 (does VMD-MFGNN differ significantly from standard baselines) is answered with a qualified yes that survives multiple-testing correction against the deep-learning baselines specifically, while remaining indistinguishable from ARIMA throughout — a real but narrow result that RQ0 puts in context rather than contradicts.

This paper's contribution is therefore not the architecture it set out to propose. It is the evaluation protocol — leakage-safe VMD, capacity-matched ablations, a naive baseline this literature typically omits, and a multiple-testing-corrected significance test — applied honestly enough to surface, rather than paper over, the naive-baseline result, the mechanism behind the negative ablation finding and a verified attempt to fix it, and a constant-forecast artifact that would otherwise have quietly overturned this paper's central negative finding. We offer three things we believe are useful beyond this specific architecture: a quantified, transferable diagnosis of embedding collapse relevant to other learned-graph financial GNN architectures at similarly low node counts, MBTI-Net-style designs included; a demonstration that diagnosing and fixing such a collapse mechanically does not guarantee the fix produces a learning signal, which we believe generalizes to other low-signal auxiliary parameters trained jointly with a dominant pathway; and a concrete, quantified illustration of why directional accuracy must be reported alongside prediction sign concentration and a majority-class null, since a degenerate model can otherwise report a plausible-looking accuracy figure by accident. A study that tests a hypothesis rigorously, explains why it failed, tests its own explanation, and reports all of it plainly, including a retraction of its own promising-looking intermediate result, is a more useful contribution to this literature than one that reports only confirming evidence.

Future work includes: (1) an explicit sparsity or entropy regularizer on the adjacency, or an auxiliary loss that directly rewards differentiated per-band structure, since Section V-F shows that removing weight decay's pull alone is insufficient—the task gradient reaching these embeddings needs to be strengthened or supplemented, not merely left unopposed; (2) a walk-forward evaluation protocol with multiple folds, enabling paired significance testing on the ablations; (3) extending multi-seed replication beyond the two RQ1 variants covered in Section V-G to the full main and ablation tables, which would also let a majority-class-null and sign-concentration check (Section V-H) be applied systematically across seeds rather than only where degeneracy happened to be noticed, and would put the tables' point estimates on the same footing as the 7.54% replicate noise floor that section measures; (4) symmetric hyperparameter optimization across all compared models, budget permitting, the wider single-model search of Section V-G having addressed only the proposed model's own search width; (5) extending the naive-baseline and economic-significance checks (Section V-J) to other base-metal commodities to test whether the difficulty of beating a trivial forecast is specific to copper's 2022–2025 regime or a more general property of this problem class;

and (6) testing whether $N = 8$ nodes is simply too small a graph for this class of learned bilinear adjacency to have a meaningful task gradient at all, independent of any decay or normalization fix, by repeating the diagnosis-and-fix protocol of Section V-F on a larger multi-commodity node set.

Data Availability

All data used in this study is publicly available via the Yahoo Finance API (tickers listed in Section IV). No FRED, BDI, TC/RC, or COT data are used.

Code Availability

The full training, evaluation, diagnostic and robustness-experiment code for this study is maintained in a single repository. [TODO: insert the public code repository URL here once the repository is made public.]

CRediT Author Statement

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